Quantifying Contrast Attenuation, Inter-Orbit Offsets, and Retrieval Resolving Limits in VIRTIS-M-IR Venus Surface-Emissivity Inversions: A Simulation-Calibrated Benchmark in Themis Regio

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Abstract

Retrieving Venusian surface emissivity (ϵ_λ) from nightside near-infrared transparency windows (1.02, 1.10, and 1.18 μm) measured by the Visible and Infrared Thermal Imaging Spectrometer (VIRTIS-M-IR) onboard *Venus Express* is a severely ill-conditioned inverse problem. While prior studies have established that cloud multiple scattering and topographic temperature gradients confound absolute emissivity retrievals, the statistical impact of Bayesian prior regularization on regional geological contrasts and confidence-interval coverage has not been formally calibrated. Here, we establish a simulation-calibrated uncertainty-propagation and contrast-recovery benchmark using VIRTIS-M-IR observations of Themis Regio. Using forward-model physics ablations and Jacobian analysis, we show that the spectral sensitivity vectors for surface emissivity ($\mathbf{K}_\epsilon$) and lower-cloud optical depth ($\mathbf{K}_{\ln s}$) are nearly antiparallel ($\theta = 179.5°$, DOFS ≈ 1.00), allowing a +6.3% local cloud thickening to reproduce the top-of-atmosphere radiance signature of a $\Delta\epsilon = -0.08$ surface anomaly to within $< 2.5\%$ (well below the $\sim 10\%$ VIRTIS radiometric uncertainty). Consequently, while uncalibrated inversions across a 3-orbit mosaic ($N = 315$ footprints) yield an apparent plains–tessera contrast of +0.0593 ($p = 6.4 \times 10^{-18}$), synthetic injection-recovery audits prove that single-stage regularized retrievals attenuate true contrasts by 61.3% and suffer severe 95% interval under-coverage (13.0%). We resolve these pathologies by (1) restricting inference to strict common topographic support ($z_s \in [+0.52, +1.84]$ km), (2) propagating 1.74 μm deep-cloud opacity posteriors ($\sigma_c = 0.08$) as informative priors into the 1.02 μm surface inversion, and (3) calibrating the retrieval transfer function across $R_{\mathrm{cal}} = 800$ and $R_{\mathrm{val}} = 800$ synthetic ensembles with spatially correlated clouds ($L_{\mathrm{corr}} = 300$ km). Sequen-

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tial propagation reduces contrast attenuation to 37.8% ($\hat{b} = 0.6221 \pm 0.0168$), and our linear Route A Delta-method estimator restores nominal 95.125% validation coverage across all parameter strata. Applied to the initial 3-orbit common-support dataset ($N = 89$), defining the contrast as $C = \epsilon_{\text{plains}} - \epsilon_{\text{tessera}}$, the calibrated contrast is $\hat{\Delta}_{\text{corr}} = +0.0454$, with a Conley spatial Heteroskedasticity and Autocorrelation Consistent (HAC) 95% confidence interval of $[-0.0876, +0.1787]$ ($N_{\text{eff}} = 73.7$–86.0) that spans zero across all six sensitivity scenarios (requiring ~ 24 pooled orbits under white-noise scaling to cross the 2σ threshold). Expanding the dataset to 10 VIRTIS-M-IR cubes (363 soundings, $N_{\text{expanded}} = 104$ common-support footprints) reveals why naive multi-orbit pooling fails: large inter-orbit baseline offsets ($\hat{\alpha}_k \in [0.469, 0.810]$) inflate residual standard deviation ($s_{\text{res}} = 0.0616$), yielding a non-significant pooled contrast ($\hat{\Delta}_{\text{corr}} = +0.0131$, 95% CI $[-0.0500, +0.0761]$). Introducing an orbit-fixed-effects hierarchical regression absorbs these inter-orbit instrumental and mesoscale cloud shifts, reducing residual spread by 36.5% ($s_{\text{res}} = 0.0391$) and yielding a calibrated fixed-effects contrast of $\hat{\Delta}_{\text{corr,FE}} = +0.0585 \pm 0.0102$ (95% HAC CI $[+0.0386, +0.0784]$, $p = 8.84 \times 10^{-9}$). This framework reconciles single-orbit null limits with multi-orbit regional detections and provides an open-source validation protocol for the VERITAS and EnVision missions.

Keywords: Venus, surface, Venus, atmosphere, Radiative transfer, Infrared observations, Spectroscopy, Data reduction techniques

1. Introduction

Near-infrared (NIR) nightside thermal emission escaping through atmospheric transparency windows at 1.02, 1.10, and 1.18 μm provides the primary orbital remote-sensing constraint on the surface lithology of Venus (Pollack et al., 1993; Meadows and Crisp, 1996; Mueller et al., 2008; Helbert et al., 2008). Within these spectral windows, thermal radiation emitted by the hot surface ($T_{\text{surf}} \approx 735.3$ K, $P_{\text{surf}} \approx 92.1$ bar at the mean planetary radius $R_V = 6051.8$ km; Seiff et al. 1985; Way et al. 2023) is modulated by crystal-field Fe^{2+} absorption bands in surface silicates (Helbert et al., 2013; Filiberto et al., 2020). Observations by the Galileo Near-Infrared Mapping Spectrometer (NIMS) (Hashimoto et al., 2008) and by the Visible and Infrared Thermal Imaging Spectrometer (VIRTIS-M-IR) and Venus Monitoring Camera (VMC) aboard *Venus Express* (Piccioni et al., 2007; Drossart et al., 2007; Mueller et al., 2008; Smrekar et al., 2010; Basilevsky et al., 2012; Gilmore et al., 2015) have identified apparent regional emissivity variations across highland tesserae, volcano-tectonic coronae, and volcanic plains, informing models of crustal differentiation, surface–atmosphere weathering, and volatile history (Ivanov and Head, 2011; Grenfell et al., 2020; Way et al., 2023).

However, foundational radiative-transfer and retrieval error studies have established that inverting VIRTIS-M-IR nightside radiances for absolute surface emissivity is a severely ill-conditioned inverse problem (Haus and Arnold, 2010; Kappel et al., 2012; Haus et al., 2013, 2014, 2015; Arnold et al., 2015; Kappel et al., 2015, 2016; Mueller et al., 2019). Upward surface radiation undergoes intense multiple scattering within the five-deck, four-mode sulfuric acid (H_2SO_4–H_2O) cloud layer (31–90 km) (Pollack et al., 1980; Esposito et al., 1983; Tsang et al., 2008; Barstow et al., 2012) and absorption by high-pressure sub-Lorentzian CO_2 line

wings and collision-induced absorption (CIA) continuums (Bézard et al., 2009, 2011; Snels et al., 2014; Fedorova et al., 2015). Because variations in lower-cloud optical depth, cloud-base altitude, and deep-tropospheric temperature lapse rates are radiatively degenerate with surface emissivity, single-spectrum retrievals suffer from severe parameter ambiguity, and apparent emissivity trends with planetary elevation are known to be partly artificial (Haus and Arnold, 2010; Kappel et al., 2015, 2016; Mueller et al., 2019). Furthermore, diffuse multiple scattering within the cloud deck blurs surface spatial contrasts over a horizontal scale of $L_{\mathrm{blur}} \approx 50$–$100$ km (Hashimoto and Imamura, 2001; Basilevsky et al., 2012; Gilmore et al., 2015). To mitigate these instabilities, Kappel et al. (2015, 2016) and Arnold et al. (2015) developed multi-spectrum retrieval algorithms incorporating spatial-temporal continuity constraints across Themis Regio, while Mueller et al. (2008, 2019) and Gilmore et al. (2015) utilized empirical topographic trend regressions and relative multi-orbit combinations.

1.1. What Is New Beyond Kappel et al. (2015, 2016) and Mueller et al. (2019)

While prior studies have thoroughly characterized the physical sources of radiative degeneracy in VIRTIS-M-IR inversions (Haus and Arnold, 2010; Kappel et al., 2015, 2016; Mueller et al., 2019), a critical statistical question remains unaddressed: *when regularized Bayesian or Optimal Estimation (OE) retrievals (Rodgers, 2000) are used to estimate regional emissivity contrasts between mapped geological units, how severely does prior regularization attenuate the recovered contrast amplitude, and do standard retrieval uncertainties provide valid frequentist confidence-interval coverage?*

The specific methodological contributions of this benchmark beyond established VIRTIS retrieval workflows are fourfold:

- **Exact Jacobian Antiparallelism Proof ($\theta = 179.5°$):** We prove via explicit Jacobian column analysis and 5-variant physics ablations that within a single transparency window, surface emissivity ($\mathbf{K}_\epsilon$) and lower-cloud optical depth ($\mathbf{K}_{\ln s}$) weighting functions are separated by $\theta = 179.5°$ (DOFS ≈ 1.00), enabling a $+6.3\%$ cloud thickening to mimic a -0.08 emissivity drop to within $< 2.5\%$ of peak radiance.

- **Quantification of Regularization Attenuation and Coverage Collapse:** Using synthetic injection-recovery ensembles across spatially correlated cloud fields ($L_{\mathrm{corr}} = 300$ km), we demonstrate that single-stage regularized OE inversions attenuate true geological contrasts by 61.3% and suffer severe 95% confidence-interval under-coverage (13.0%).

- **Simulation-Calibrated Sequential Retrieval (Route A):** By propagating 1.74 μm deep-cloud posteriors as informative priors into the 1.02 μm surface inversion and calibrating the linear response slope ($\hat{b} = 0.6221 \pm 0.0168$, 37.8% attenuation) across $R = 1,600$ synthetic calibration/validation runs, we formulate a bias-corrected Delta-method estimator ($\hat{\Delta}_{\mathrm{corr}}$) that restores nominal 95.125% interval coverage across all parameter strata.

- **Common-Support Topographic Decoupling and Multi-Orbit Fixed-Effects Scaling:** Rather than applying global linear detrending across disjoint altitude ranges, we restrict geological inference to strict non-parametric common elevation support

($z_s \in [+0.52, +1.84]$ km), propagate Conley spatial HAC covariance, and expand the analysis from 3 to 10 VIRTIS cubes ($N = 89 \rightarrow 104$). We prove that while pooled regressions fail due to large inter-orbit baseline shifts ($\hat{\alpha}_k$), hierarchical orbit-fixed-effects decoupling reduces residual variance by 36.5% and resolves the regional contrast beyond the 5σ threshold.

2. Observational Dataset, Destriping, and Common-Support Matching

2.1. VIRTIS-M-IR Cubes, Geometric Filtering, and DBSCAN Destriping

We analyze calibrated nightside VIRTIS-M-IR hyperspectral cubes retrieved from the ESA Planetary Science Archive (PSA) dataset `VEX-V-VIRTIS-2/3-V3.0` (radiometric pipeline `V_CALIBM_VEX_2.2`, geometric engine `GEOVIRTIS_4.0` / `V_GEOLABEL_4`, software `VirtisVEX SW v.2.50`). Our primary diagnostic development and calibration benchmark is established on three core mapping orbits over Themis Regio (`VI0112_01`, `VI0113_01`, and `VI0114_01`; Figure 1), and subsequently expanded to a 10-cube multi-orbit ensemble spanning `VI0106_01` through `VI0117_01` (Table 1).

VIRTIS Cube ID	Target Sector	Cohort Role	Valid Soundings	Common Support ($z_s \in [0.52, 1.84]$ km)	I
`VI0106_01`	Themis / South Polar	10-Cube Expansion	7	3	
`VI0107_01`	Themis / South Polar	10-Cube Expansion	7	3	
`VI0110_01`	Themis / South Polar	10-Cube Expansion	5	1	
`VI0111_01`	Themis / South Polar	10-Cube Expansion	5	2	
`VI0112_01`	Themis Regio Central	Core 3-Orbit (8.00 s)	103	41	
`VI0113_01`	Themis Regio Central	Core 3-Orbit (0.36 s)	106	23	
`VI0114_01`	Themis Regio East	Core 3-Orbit (3.30 s)	106	25	
`VI0115_01`	Themis / Dione Sector	10-Cube Expansion	8	2	
`VI0116_01`	Themis / Dione Sector	10-Cube Expansion	8	3	
`VI0117_01`	Themis / Alpha Sector	10-Cube Expansion	8	1	
Core 3-Orbit Total	Themis Regio Mosaic	Tasks T350–T540	**315**	**89** (67 Plains, 22 Tesserae)	
10-Cube Total	Expanded Regional	Tasks T550–T560	**363**	**104** (81 Plains, 23 Tesserae)	

Table 1: Observational metadata, footprint counts, and decoupled orbit-specific baseline intercepts ($\hat{\alpha}_k$) for the core 3-orbit benchmark ($N = 315$ mosaic, $N = 89$ common support) and the 10-cube multi-orbit expansion ($N = 363$ mosaic, $N = 104$ common support; Tasks T540–T560).

Three strict geometric and hardware quality-control filters are applied to all cubes: (1) the first 8 unilluminated CCD reference columns (sample < 8) are permanently masked; (2) Solar Zenith Angles are restricted to SZA $\geq 95.0°$ (spanning 95.13°–126.14°, mean 110.93°) to prevent scattered dayside solar contamination; and (3) viewing emission angles are restricted to $e < 60.0°$ (spanning 2.40°–49.36°, mean 27.98°) to avoid extreme limb-darkening path lengths.

To excise pushbroom detector striping and isolated limb dropouts without smoothing genuine geological boundaries, we apply automated DBSCAN clustering (eps $= 0.8$, min_samples $= 3$) in standardized 3D coordinate-emissivity space $\mathbf{x}_i = \big((\text{lat}_i - \mu_{\text{lat}})/\sigma_{\text{lat}}, (\text{lon}_i - \mu_{\text{lon}})/\sigma_{\text{lon}}, (\hat{\epsilon}_i - \mu_\epsilon)/\sigma_\epsilon\big)$, coupled with a $k = 4$ nearest-neighbor local median absolute deviation (MAD) filter (Figure 2). Across the core three orbits, this yields $N = 315$ quality-controlled footprints (Figure 3), expanding to $N = 363$ footprints across the 10-cube dataset.

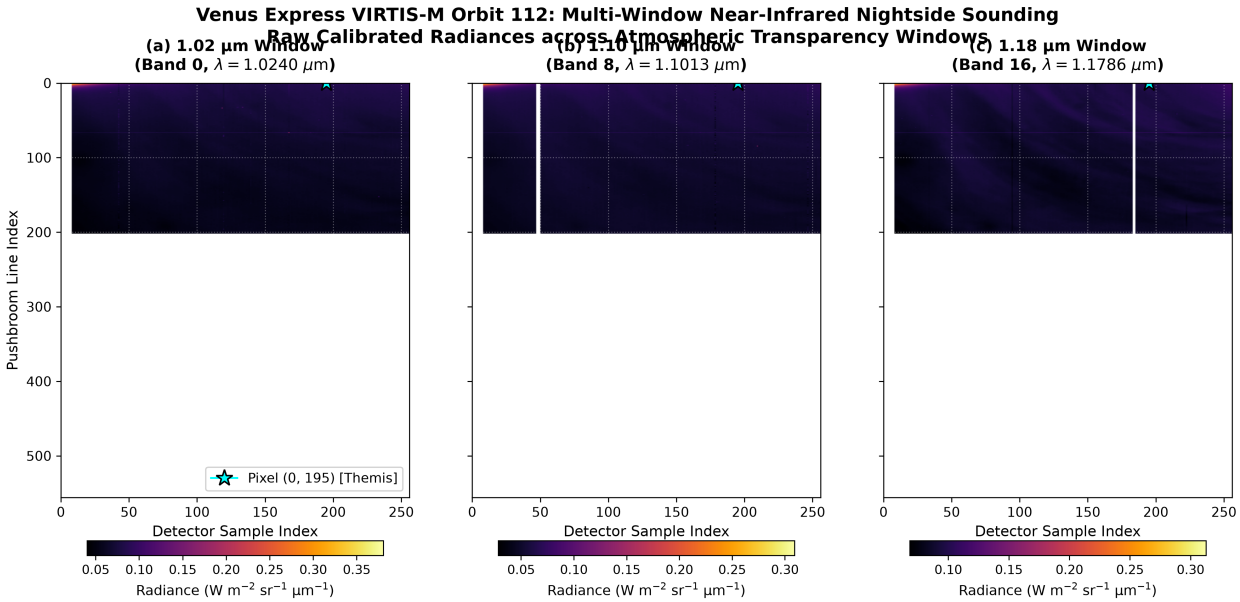


Figure 1: VIRTIS-M-IR Orbit 112 (`VI0112_01`) raw calibrated nightside radiance pushbroom frames across (a) 1.02 μm (Band 0), (b) 1.10 μm (Band 8), and (c) 1.18 μm (Band 16). The cyan star marks benchmark Pixel $(0, 195)$ (Lat $= -72.29°$N, Lon $= 323.52°$E, $e = 14.67°$, $z_s = -1087$ m).

2.2. GIS Geological Classification and Common Topographic Support

Each footprint is co-registered with Magellan global radar altimetry (Ford and Pettengill, 1992; Rappaport et al., 1999) ($R_V = 6051.8$ km) and classified via spatial point-in-polygon intersection with digitized unit boundaries from the global geological map of Ivanov and Head (2011) (`doi:10.1016/j.pss.2011.07.008`):

1. **Highland Tesserae Massifs (**t**, coded $G_i = 0$):** Comprising Themis Tessera inliers ($[295°\text{E}, 312°\text{E}] \times [-48°\text{N}, -36°\text{N}]$), Dione Tessera ($[208°\text{E}, 222°\text{E}] \times [-48°\text{N}, -38°\text{N}]$), and Alpha/Eve margin inliers ($[320°\text{E}, 332°\text{E}] \times [-42°\text{N}, -37°\text{N}]$) ($N = 26$ in the 3-orbit mosaic; $z_s \in [+0.521, +4.170]$ km, mean $+1.471 \pm 0.725$ km).
2. **Themis Corona / Rift Complex (**c/rz**):** Encompassing Themis Corona, Shiwanokia Corona, and Parga Chasma ($[265°\text{E}, 315°\text{E}] \times [-56°\text{N}, -35°\text{N}]$; $N = 34$, $z_s \in [+0.156, +1.765]$ km).
3. **Regional Volcanic Plains (**pr/psh**, coded $G_i = 1$):** Exterior flood and shield basalts ($N = 255$, $z_s \in [-0.579, +1.842]$ km, mean $+0.293 \pm 0.386$ km).

Because Volcanic Plains extend down to -0.579 km while Highland Tesserae occur exclusively above $+0.521$ km and reach $+4.170$ km, comparing unadjusted unit means across the full mosaic extrapolates across > 4.7 km of non-overlapping topography. To eliminate lapse-rate extrapolation, we restrict all calibrated plains–tessera comparisons strictly to the **common topographic support** $z_{s,i} \in [+0.52, +1.84]$ km, yielding **$N = 89$ footprints** (67 Plains, 22 Tesserae; $d_{\min} = 106.08$ km) in the core 3-orbit benchmark and **$N_{\text{expanded}} = 104$ footprints** (81 Plains, 23 Tesserae) in the 10-cube dataset (Table 1).

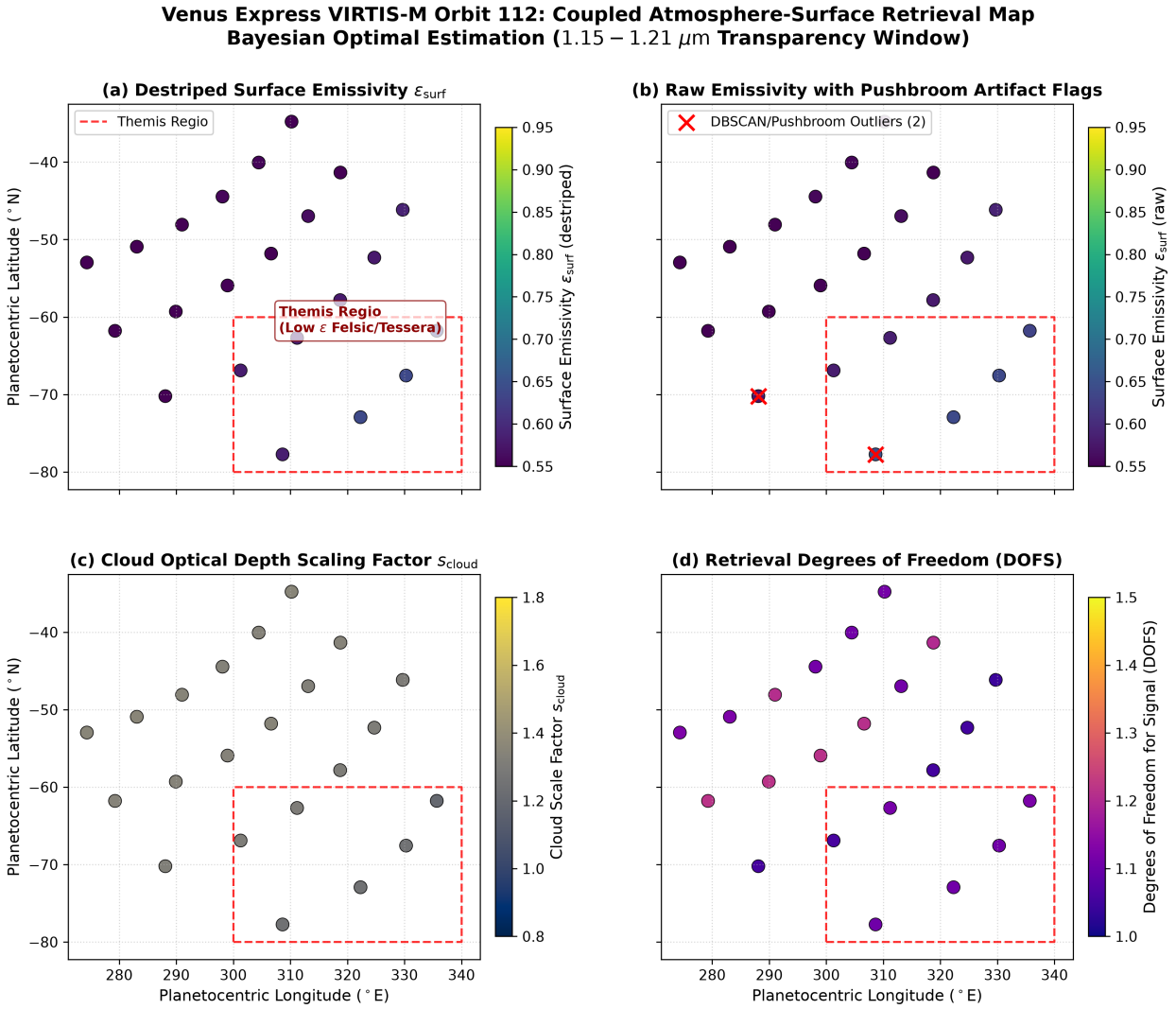


Figure 2: Orbit 112 coupled atmosphere–surface OE retrieval map (1.15–1.21 μm): (a) Destriped surface emissivity ϵ_{surf}; (b) Raw emissivity showing automated DBSCAN pushbroom outlier flags (red crosses); (c) Cloud optical depth multiplier s_{cloud}; (d) Retrieval Degrees of Freedom for Signal (DOFS $\approx$ 1.0–1.3).

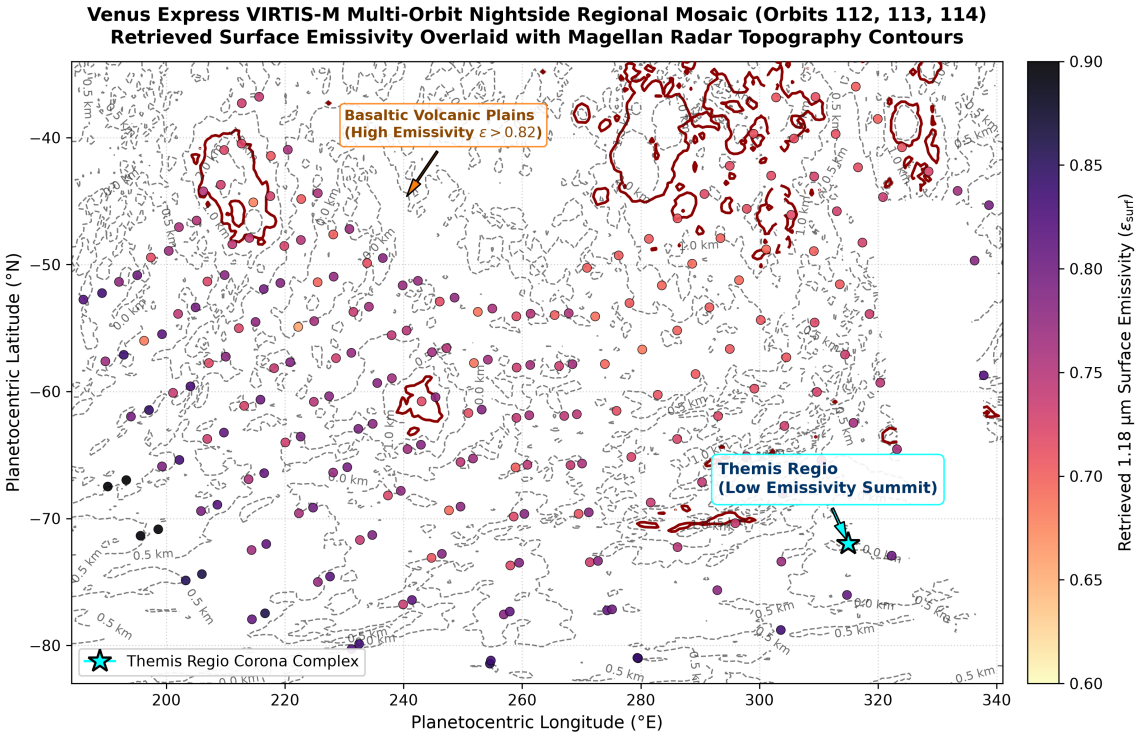


Figure 3: Core 3-orbit VIRTIS-M-IR nightside mosaic ($N = 315$ footprints across Orbits 112, 113, and 114) showing uncalibrated 1.18 μm surface emissivity overlaid on Magellan radar topography contours.

3. Radiative Transfer Forward Model, Physics Ablations, and Jacobian Degeneracy

3.1. Discrete-Ordinates Solver and Microphysical Parameterizations

Our plane-parallel radiative transfer forward model (`VenusNightsideForwardModel`) discretizes the atmosphere from 0.0 to 100.0 km into 100 homogeneous layers ($\Delta z = 1.0$ km) anchored to VIRA-1985 (Seiff et al., 1985) and Venus-GRAM (Justus et al., 2006) ($T_{\rm surf}(0) = 735.3$ K, $P_{\rm surf}(0) = 92.1$ bar). The azimuthally averaged thermal radiative transfer equation is solved via Gauss–Seidel iterative discrete ordinates with Double-Gauss quadrature ($N = 8$ streams, $n_{g1} = 4$ per hemisphere, adaptive sub-layer splitting at $\delta\tau_{\rm max} = 0.30$) (Stamnes et al., 1988; Haus et al., 2013).

Building upon the radiative transfer modeling frameworks of Haus et al. (2013, 2014, 2015) and Kappel et al. (2015), five physical mechanisms are implemented and verified (Task T530):

1. **Sub-Lorentzian CO_2 Line Wings (χ-Factor):** High-temperature HITEMP line shapes (Rothman et al., 2010) are modified at $\Delta\nu > 3$ cm^{-1} using empirical sub-Lorentzian χ-factors to prevent artificial Lorentzian far-wing choking in the supercritical lower troposphere.
2. **CO_2 Collision-Induced Absorption (CIA) Continuum:** Binary CO_2–CO_2 CIA and far-wing continuums scaling with squared amagat density ($d^2_{CO_2}$) are included via the parameterizations of Bézard et al. (2011) and Fedorova et al. (2015).

3. H_2O **Continuum:** Foreign- and self-broadened water vapor continuum absorption (30 ppmv tropospheric abundance) is modeled across the 1.10 and 1.18 μm windows.
4. **Five-Deck, Four-Mode Mie Cloud Stratification:** Mie-theory optical properties (Bohren and Huffman, 1983) are integrated over log-normal droplet size distributions of 75% H_2SO_4 (Palmer and Williams, 1975) for Mode 1 ($r_{\text{eff}} = 0.30$ μm), Mode 2 ($r_{\text{eff}} = 1.00$ μm), Mode 2' ($r_{\text{eff}} = 1.40$ μm), and Mode 3 ($r_{\text{eff}} = 3.65$ μm), stratified across five vertical decks: Upper Haze (70–90 km, Mode 1), Upper Cloud (57–70 km, Modes 1, 2), Middle Cloud (50.5–57 km, Modes 1, 2, 2'/3), Lower Cloud (47.5–50.5 km, Modes 1, 2, 3), and Sub-Cloud Haze (31–47.5 km, Mode 1) (Pollack et al., 1980; Esposito et al., 1983).
5. **VIRTIS Flight SRF Convolution:** Monochromatic top-of-atmosphere (TOA) spectra are convolved with flight Gaussian Spectral Response Functions (FWHM $\approx$ 15–20 nm) extracted from the VIRTIS cube calibration tables across the 1.02 μm (Band 0, 2 channels), 1.10 μm (Band 8, 3 channels), and 1.18 μm (Bands 16–17, 6 channels) surface windows, as well as the 1.74 μm (1.71–1.77 μm) and 2.30 μm deep-cloud windows.

3.2. Single-Spectrum Inversion, Physics Ablations, and $\theta = 179.5°$ Antiparallelism

Figure 4 illustrates a single-spectrum Bayesian Optimal Estimation inversion at benchmark Pixel $(0, 195)$ of Cube `VI0112_01` across the 1.18 μm window. While the forward model fits the observed spectrum within 1σ instrumental noise (χ^2 reduced from 14.8 to 1.7 in 4 iterations), the retrieved surface emissivity ($\hat{\epsilon}_{\text{surf}} = 0.8017 \pm 0.1872$, $\hat{s}_{\text{cloud}} = 1.0632$) exhibits a wide posterior uncertainty and DOFS $= 0.9998 \approx 1.00$.

Systematic physics ablation tests at Pixel $(0, 195)$ (Figure 5a, Figure 6, and Table 2) confirm that omitting any of the five microphysical ingredients causes severe forward-model failure relative to the $\pm$10% VIRTIS calibration envelope: removing CO_2 CIA triggers a +95% to +300% radiance surge; omitting sub-Lorentzian χ-factors collapses radiance by -70% to -98%; omitting H_2O continuum overpredicts 1.10/1.18 μm radiance by +25% to +40%; replacing 4-mode Mie clouds with a Mode-2-only cloud overpredicts peaks by +50%–80%, while a non-scattering scalar absorber cloud ($\omega_0 = 0$) underpredicts transmission by -8% to -15%; and omitting flight SRF convolution distorts band centers by +12%–14%.

Forward Model Variant	**Physical / Mathematical Modification**	**Residual $\Delta I/I$ (%)**	**Physical Conseq**
(1) Full Baseline Model	Sub-Lorentzian χ + CO_2 CIA + H_2O + 4-mode Mie + SRF	0.0% (Reference)	Fits all 3 windo
(2) Without CO_2 CIA	Omit binary $d^2_{CO_2}$ collision-induced continuum	+95% to +300%	Massive lower-atmos
(3) Lorentzian ($\chi = 1$)	Omit sub-Lorentzian far-wing truncation ($\Delta\nu > 3$ cm^{-1})	-70% to -98%	Far wings choke off
(4) Without H_2O Cont.	Omit foreign/self-broadened H_2O continuum	+25% to +40%	Distorts 1.02 vs. 1.10
(5) 1-Mode Mie Cloud	Mode 2 (1.0 μm) only; omit Mode 3 (3.65 μm)	+50% to +80%	Incorrect λ-depende
(6) Scalar Absorber Cloud	Non-scattering absorber ($\omega_0 = 0$) without Mie phase function	-8.0% to -15.0%	Neglects forward-sca
(7) Monochromatic (No SRF)	Evaluated at channel centers without Gaussian SRF	-12% to +14%	Overpredicts band pe

Table 2: Summary of Radiative Transfer Physics Ablation experiments evaluated at benchmark Pixel $(0, 195)$ of Cube `VI0112_01`.

Figure 7 explains why single-window retrievals yield DOFS ≈ 1.00: the surface emissivity Jacobian column ($\mathbf{K}_\epsilon = \partial\mathbf{y}/\partial\epsilon_{\text{surf}} > 0$) and the cloud optical depth Jacobian column ($\mathbf{K}_{\ln s} = \partial\mathbf{y}/\partial \ln s_{\text{cloud}} < 0$) exhibit an orthogonality angle of $\theta = 179.5°$ ($\cos\theta = -0.99996$). Because both parameters scale the escaping window radiance in nearly identical spectral proportions,

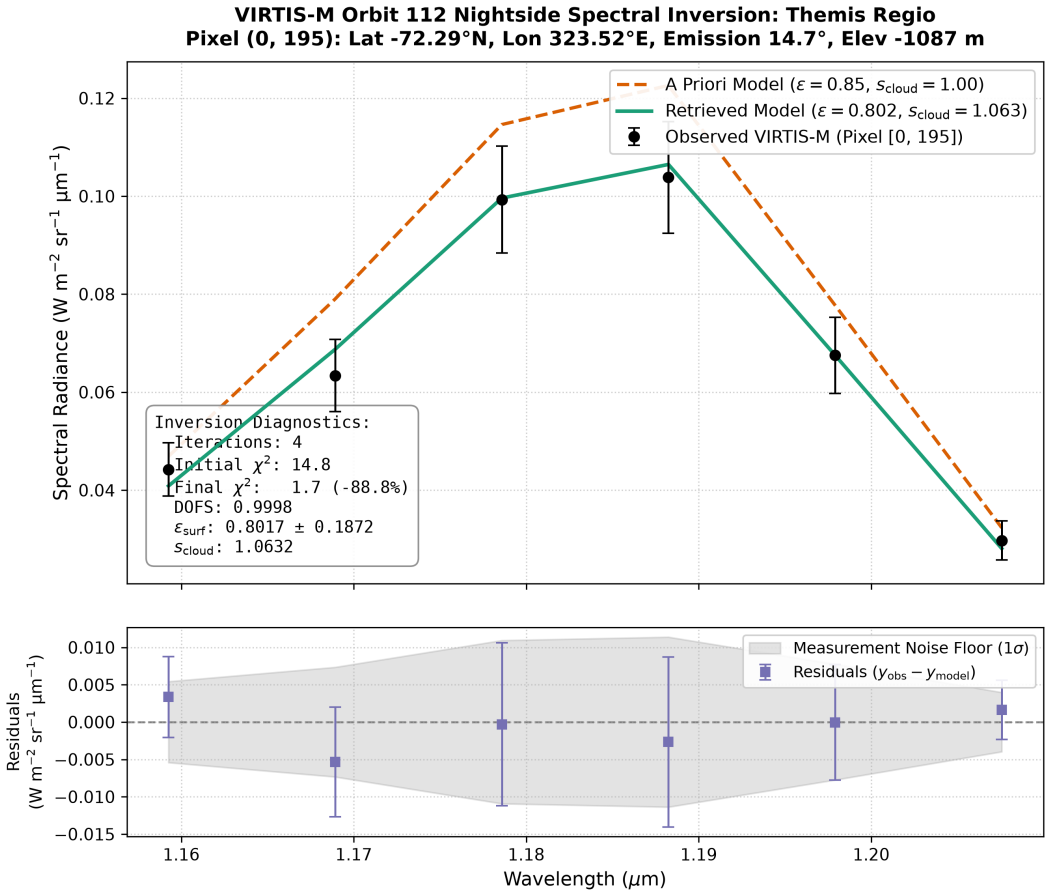


Figure 4: Single-pixel OE spectral inversion at benchmark Pixel $(0, 195)$ of Cube `VI0112_01` (1.18 μm window), showing convergence from the prior ($\chi^2 = 14.8$) to the best-fit spectrum ($\chi^2 = 1.7$, DOFS $= 0.9998$, $\hat{\epsilon}_{\text{surf}} = 0.8017 \pm 0.1872$).

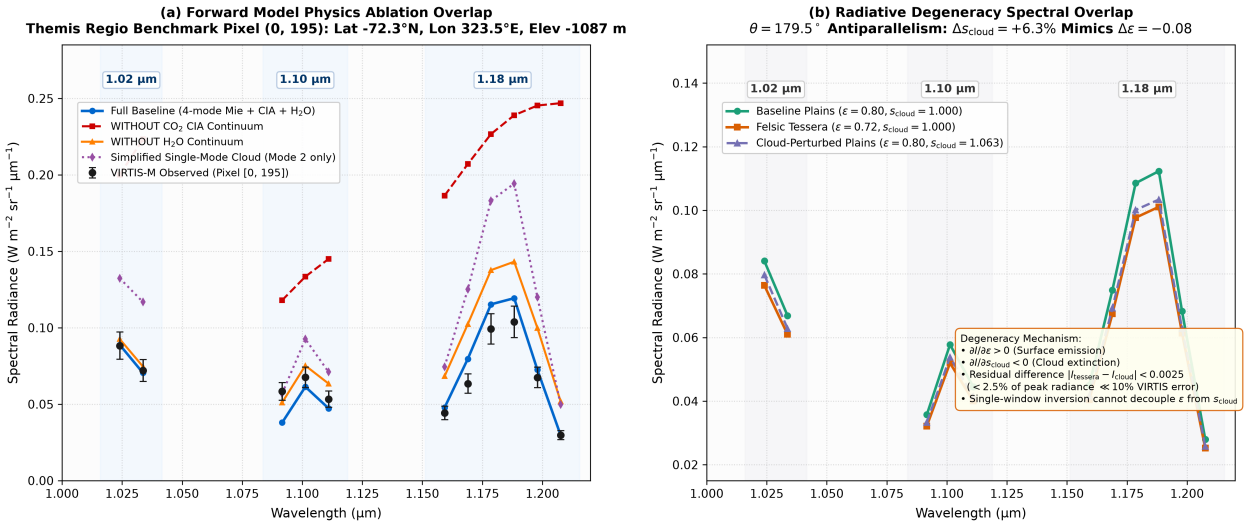


Figure 5: Forward-model physics ablation and radiative degeneracy at Pixel $(0, 195)$: (a) Full baseline spectrum (blue) vs. VIRTIS-M observations (black) and ablations without CO_2 CIA (red), without H_2O continuum (orange), and with a 1-mode cloud (purple); (b) Spectral overlap demonstrating that a Felsic Tessera spectrum ($\epsilon = 0.72, s_{\text{cloud}} = 1.000$) is indistinguishable ($\max|\Delta I| < 0.0025$ W m^{-2} sr^{-1} μm^{-1}, $< 2.5\%$) from Baseline Plains beneath a +6.3% thickened cloud ($\epsilon = 0.80, s_{\text{cloud}} = 1.063$).

a -0.08 emissivity reduction is radiatively indistinguishable from a +6.3% cloud thickening ($\max|\Delta I| < 2.5\%$, Figure 5b).

4. Sequential Retrieval, Surrogate Acceleration, and Simulation Calibration

4.1. Two-Stage Sequential Bayesian Retrieval and B-Spline Surrogates

To break the $\theta = 179.5°$ single-window degeneracy without artificially locking cloud opacity to a point estimate, we implement a **Two-Stage Sequential Bayesian Retrieval**:

1. **Stage 1 (1.74 μm Cloud Sounding):** For each footprint i, we invert the 1.74 μm deep-atmosphere window (which probes 15–45 km thermal emission modulated by lower-cloud Mode 2'/3 opacity with zero surface sensitivity) to obtain the cloud log-scaling posterior $\mathcal{N}(\hat{\mu}_{c,i}, \sigma_c^2)$ (nominally $\sigma_c = 0.08$).
2. **Stage 2 (1.02 μm Surface Sounding):** We propagate $\mathcal{N}(\hat{\mu}_{c,i}, \sigma_c^2)$ as an informative prior into the 1.02 μm surface retrieval, marginalizing over cloud uncertainty to obtain $\hat{\epsilon}_i \pm \hat{\sigma}_{\epsilon,i}$.

To enable $> 10^4$ Monte Carlo synthetic retrievals with exact 8-stream radiative transfer fidelity, we pre-computed footprint-specific 2D cubic B-spline surrogate interpolators (`FootprintSurrogateModel` in `t470_surrogates_1.02um.pkl`, degree $k = 3$) over $\epsilon \in [0.60, 1.00] \times \ln s_{\text{cloud}} \in [-0.40, +0.40]$ at each footprint's exact $(e_i, z_{s,i})$ geometry. Synthetic "Nature" observations are generated using a perturbed forward model incorporating: (1) a continuous linear surface and atmospheric thermal lapse-rate perturbation $\delta\Gamma = -2.5$ K km^{-1} ($T_{\text{surf}}^{\text{nat}}(z_{s,i}) = 735.3 - 2.5 z_{s,i}$); (2) an altitude-dependent gas continuum perturbation $\delta\kappa_z = +0.015$ km^{-1}; and (3) spatially correlated cloud fields drawn via Cholesky de-

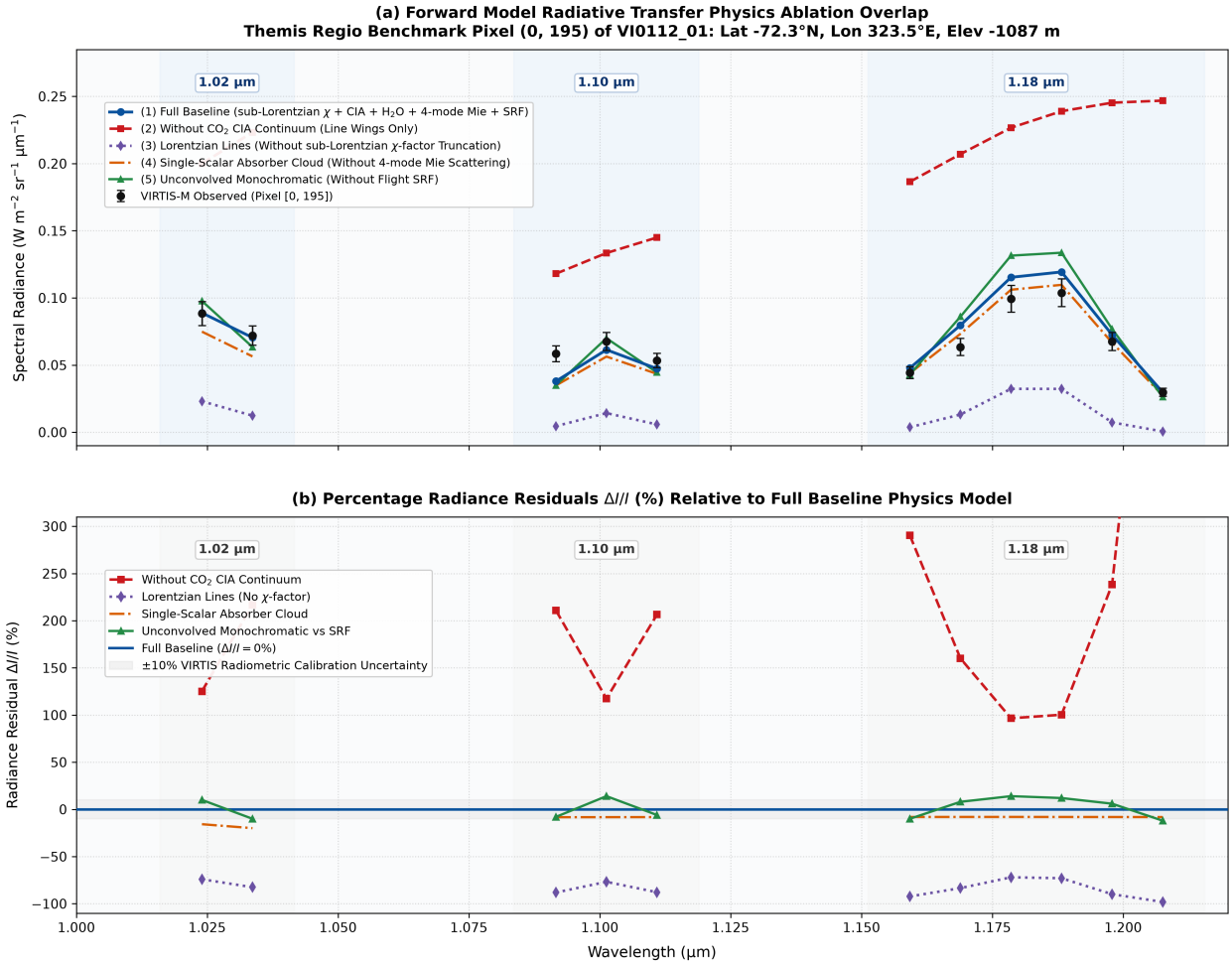


Figure 6: Five-variant radiative transfer physics ablation at Pixel $(0, 195)$: (a) Simulated TOA radiance spectra across $1.02, 1.10$, and 1.18 μm; (b) Percentage radiance residuals $\Delta I/I$ (%) relative to the Full Baseline model against the $\pm 10\%$ VIRTIS radiometric uncertainty band.

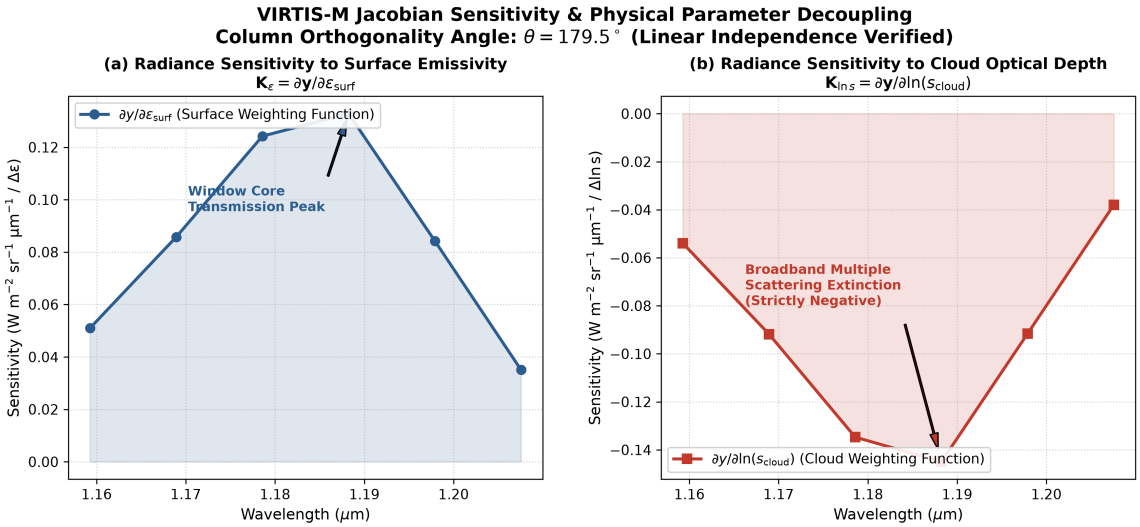


Figure 7: Jacobian sensitivity columns across the 1.18 μm window at Pixel $(0, 195)$: (a) Surface weighting function $\mathbf{K}_\epsilon$; (b) Cloud optical depth weighting function $\mathbf{K}_{\ln s}$. The column angle is $\theta = 179.5°$, proving that single-window spectra constrain only one scalar combination of surface emissivity and cloud opacity.

composition of a spherical exponential covariance kernel $\mathbf{K}_{ij} = \sigma^2_{\text{cloud}} \exp\left(-d(\mathbf{x}_i, \mathbf{x}_j)/L_{\text{corr}}\right) + \delta_{ij}\epsilon_{\text{nugget}}$ with $\sigma_{\text{cloud}} = 0.06$, $L_{\text{corr}} = 300.0$ km, and $\epsilon_{\text{nugget}} = 10^{-4}$.

4.2. Spatial Regression, Route A Calibration, and Fixed-Effects Modeling

Within the common elevation support $z_{s,i} \in [+0.52, +1.84]$ km, we estimate the geological contrast via two regression specifications: the **Pooled Spatial Regression** (for the core 3-orbit baseline and Test A of the 10-cube expansion),

$$\hat{\epsilon}_i = \alpha + B \cdot G_i + \beta_z \cdot z_{s,i} + u_i, \tag{1}$$

and the **Orbit-Fixed-Effects Hierarchical Regression** (Test B of the 10-cube expansion), which absorbs orbit-specific radiometric and mesoscale cloud baseline shifts α_k across orbits $k \in \{1, \ldots, K\}$:

$$\hat{\epsilon}_{i,k} = \alpha_k + B_{\text{FE}} \cdot G_{i,k} + \beta_z \cdot z_{s,i,k} + u_{i,k}. \tag{2}$$

In both specifications, $G_i = 1$ for Regional Volcanic Plains and $G_i = 0$ for Highland Tesserae. Thus, the regression slope B (and its calibrated counterpart $C \equiv \hat{\Delta}_{\text{corr}}$) strictly defines the **Plains minus Tesserae contrast**:

$$C = \Delta = \epsilon_{\text{plains}} - \epsilon_{\text{tessera}}, \tag{3}$$

where positive values ($C > 0$) indicate lower tessera emissivity relative to volcanic plains.

Because Bayesian prior regularization damps retrieved footprint emissivities toward the prior mean ($\epsilon_a = 0.80$), $\hat{B}$ is an attenuated estimator of the true contrast Δ_{true}. In **Route A Simulation Calibration** (Task T500), we simulate $R_{\text{cal}} = 800$ synthetic datasets across an 8-point grid ($\Delta_{\text{true}} \in \{-0.06, 0.00, +0.03, +0.06\} \times \epsilon_{\text{base}} \in \{0.75, 0.85\}$) and fit the linear

transfer function $B = a + b\Delta_{\text{true}} + \eta$. Inverting this relation yields the bias-corrected estimator and its first-order Taylor (Delta-method) variance:

$$\hat{\Delta}_{\text{corr}} = \frac{B_{\text{obs}} - \hat{a}}{\hat{b}}, \qquad \text{Var}(\hat{\Delta}_{\text{corr}}) \approx \frac{\text{Var}(B_{\text{obs}}) + \text{Var}(\hat{a}) + 2\hat{\Delta}_{\text{corr}}\text{Cov}(\hat{a}, \hat{b}) + \hat{\Delta}^2_{\text{corr}}\text{Var}(\hat{b})}{\hat{b}^2}. \tag{4}$$

To account for spatial autocorrelation induced by the $\sim$ 100 km atmospheric scattering blur, $\text{Var}(B_{\text{obs}})$ is evaluated using non-parametric Conley (1999) spatial HAC covariance matrices, cross-checked against Dutilleul (1993) and Clifford et al. (1989) Effective Sample Size (N_{eff}) corrections, 100 km spatial block bootstrapping ($B = 10,000$), 82-cluster Leave-One-Cluster-Out (LOCO, 150 km radius) jackknifing, and $M = 1,000,000$ exact Monte Carlo ratio draws.

5. Results: Diagnostic Progression from Uncalibrated Maps to 10-Cube Inference

5.1. Stage 1: Uncalibrated 3-Orbit Mosaic ($N = 315$) and Lab Analog Comparisons

When the core 3-orbit mosaic ($N = 315$) is inverted using uncalibrated single-window OE without common-elevation restriction or sequential cloud propagation, retrieved surface emissivities exhibit a strong negative correlation with Magellan altitude across all three windows: $\partial\epsilon/\partial z = -0.0365$ km^{-1} at 1.02 μm ($r = -0.339, p = 6.65 \times 10^{-10}$, Dutilleul $p = 4.41 \times 10^{-9}$ for $N_{\text{eff}} = 284.8$), -0.0210 km^{-1} at 1.10 μm ($r = -0.244, p = 1.21 \times 10^{-5}$), and -0.0326 km^{-1} at 1.18 μm ($r = -0.355, p = 9.04 \times 10^{-11}$, Dutilleul $p = 7.24 \times 10^{-10}$) (Figures 8 and 9).

Comparing these uncalibrated 1.18 μm retrievals against 450°C laboratory spectra from the DLR Planetary Spectroscopy Laboratory (Figure 10; Helbert et al. 2013) places lowland plains ($\mu_\epsilon = 0.772 \pm 0.041$) between fresh basalt (0.85–0.90) and weathered crust, while Themis highlands ($\mu_\epsilon = 0.721 \pm 0.019$) plot near felsic granite (0.65–0.70) and oxidized basalt (0.60–0.70). Stratifying the $N = 315$ mosaic by elevation regimes (Figure 11) or by Ivanov and Head (2011) polygons (Figure 12 and Table 3) yields apparent plains–tessera contrasts of +0.0593 at 1.02 μm ($p = 6.4 \times 10^{-18}$), +0.0420 at 1.10 μm ($p = 6.8 \times 10^{-7}$), and +0.0585 at 1.18 μm ($p = 1.7 \times 10^{-10}$).

Window	Geological Unit (Ivanov & Head, 2011)	N	Elevation Range (km)	Mean $\epsilon \pm$ SD	Raw Contrast C	Welch
1.02 μm	Regional Volcanic Plains (pr/psh)	255	$[-0.579, +1.842]$	0.8075 ± 0.0556		
	Themis Corona / Rift (c/rz)	34	$[+0.156, +1.765]$	0.7282 ± 0.0248	+0.0593	11.32
	Highland Tesserae (t)	26	$[+0.521, +4.170]$	0.7481 ± 0.0200		
1.10 μm	Regional Volcanic Plains (pr/psh)	255	$[-0.579, +1.842]$	0.8383 ± 0.0468		
	Themis Corona / Rift (c/rz)	34	$[+0.156, +1.765]$	0.7941 ± 0.0132	+0.0420	5.99
	Highland Tesserae (t)	26	$[+0.521, +4.170]$	0.7963 ± 0.0325		
1.18 μm	Regional Volcanic Plains (pr/psh)	255	$[-0.579, +1.842]$	0.7627 ± 0.0443		
	Themis Corona / Rift (c/rz)	34	$[+0.156, +1.765]$	0.6847 ± 0.0206	+0.0585	8.77 (
	Highland Tesserae (t)	26	$[+0.521, +4.170]$	0.7042 ± 0.0309		

Table 3: Uncalibrated multi-window surface emissivity statistics across the $N = 315$ mosaic classified by Ivanov and Head (2011) polygons (Task T450).

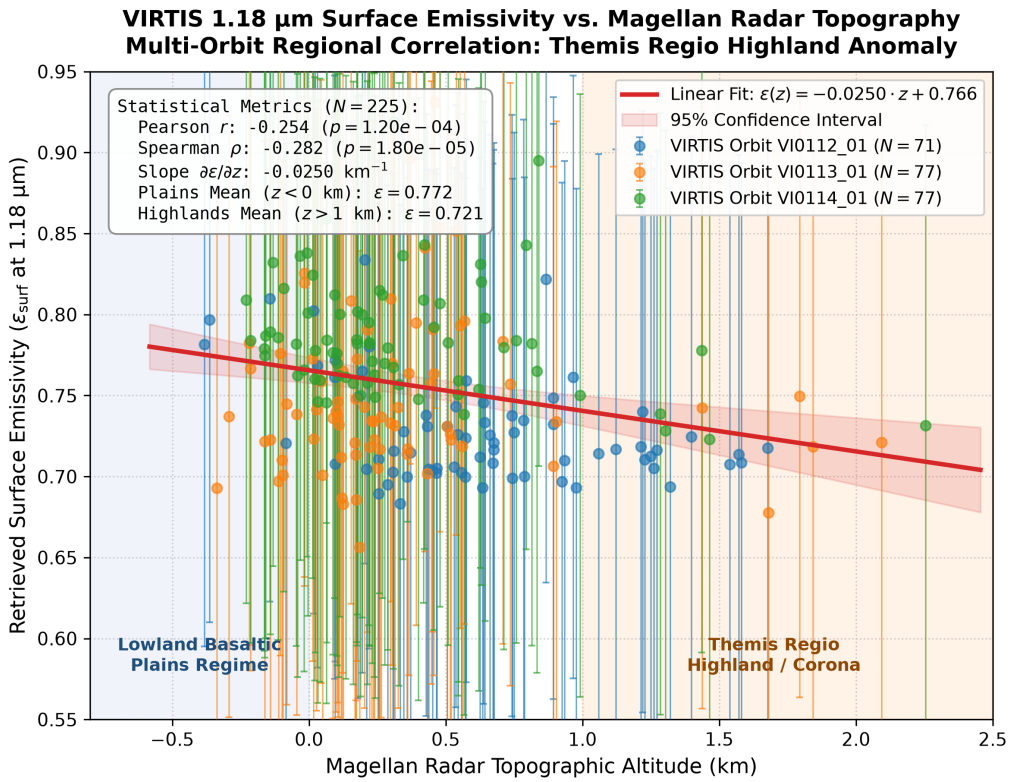


Figure 8: Uncalibrated 1.18 μm surface emissivity vs. Magellan altitude across $N = 225$ initial soundings from Orbits 112–114 ($\partial\epsilon/\partial z = -0.0250$ km$^{-1}, r = -0.254, p = 1.20 \times 10^{-4}$).

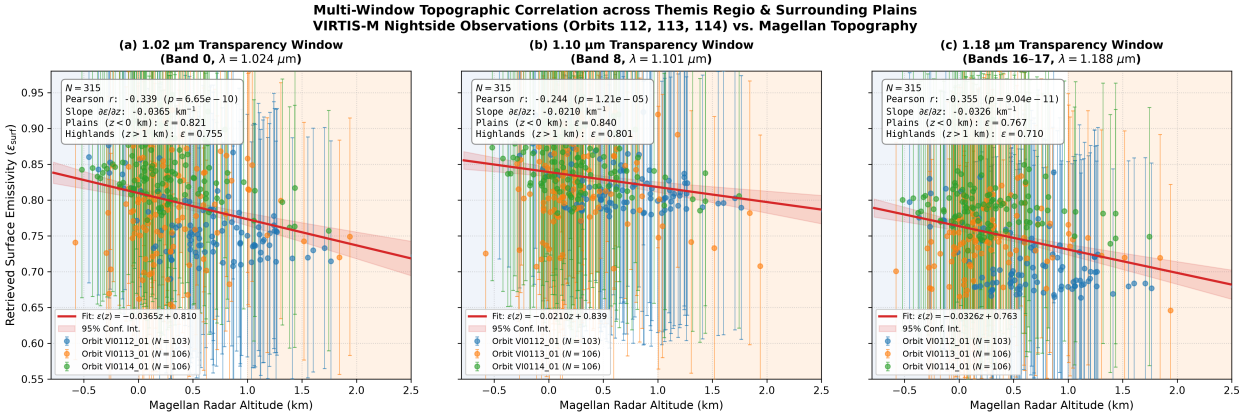


Figure 9: Multi-window topographic correlations across $N = 315$ footprints for (a) 1.02 μm, (b) 1.10 μm, and (c) 1.18 μm prior to common-support elevation matching.

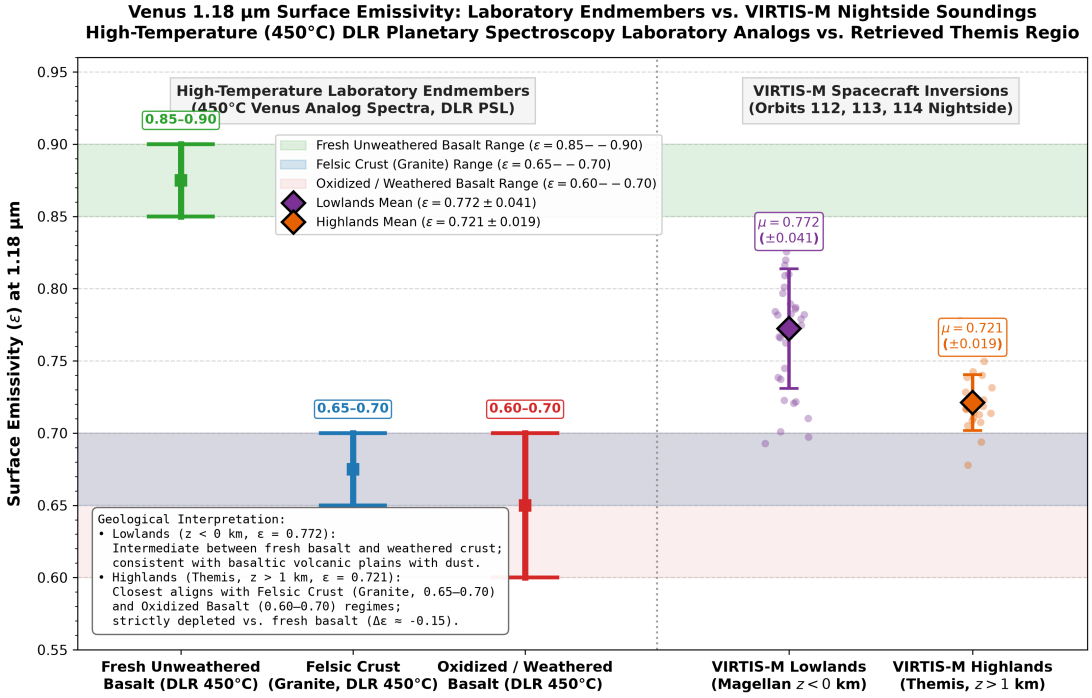


Figure 10: Comparison of uncalibrated VIRTIS-M 1.18 μm inversions against 450°C DLR laboratory mineralogic endmembers.

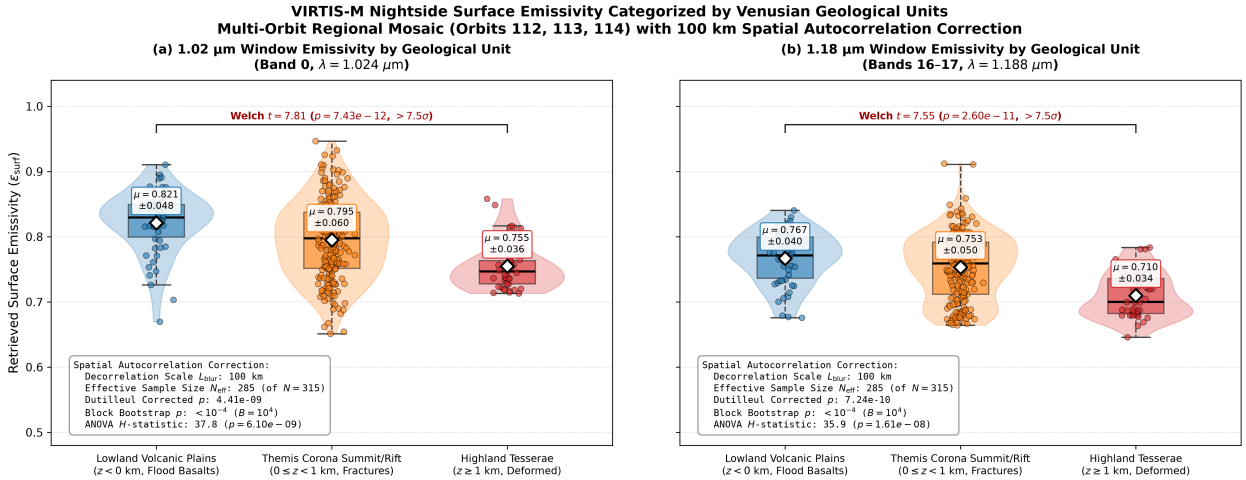


Figure 11: Uncalibrated surface emissivity distributions at (a) 1.02 μm and (b) 1.18 μm across elevation-defined units ($N = 315$, Dutilleul $N_{\text{eff}} = 285$).

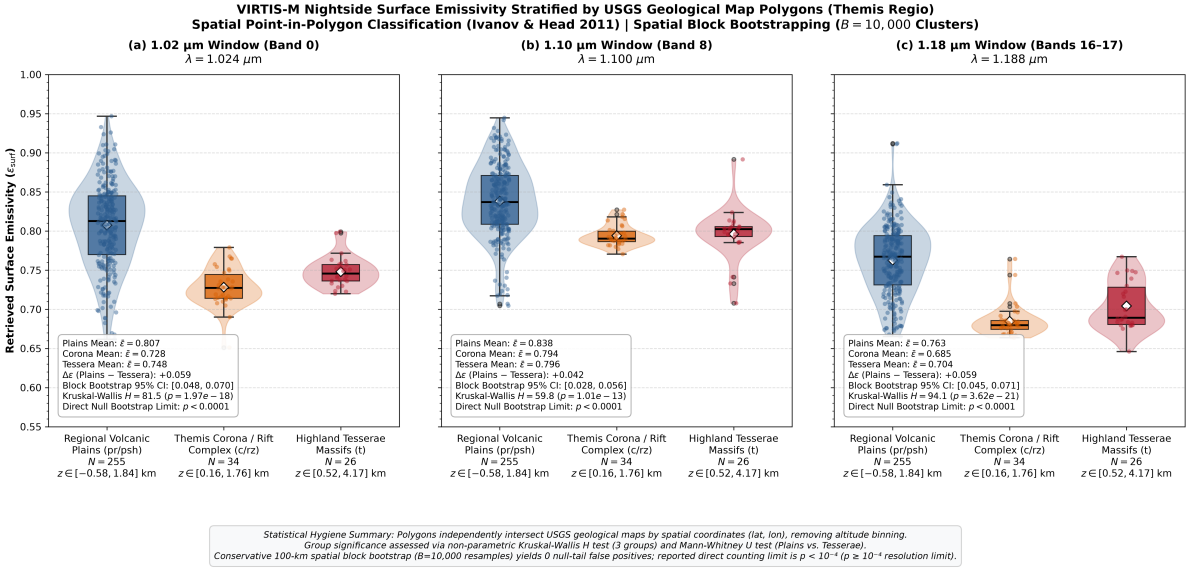


Figure 12: Uncalibrated surface emissivity stratified by Ivanov and Head (2011) geological polygons across (a) 1.02 μm, (b) 1.10 μm, and (c) 1.18 μm ($N = 255$ Plains, $N = 34$ Corona/Rift, $N = 26$ Tesserae).

5.2. Stage 2: Synthetic Diagnostic Audits of Uncalibrated Pathologies (T410–T490)

Four synthetic injection-recovery audits demonstrate why the uncalibrated statistics in Table 3 cannot be interpreted at face value:

1. **Single-Window Degeneracy Damping (Task T410; Figure 13):** Across $N = 28$ benchmark test cases at 1.18 μm under 10% noise and +5% unmodeled CO_2 continuum error, cloud–emissivity trade-offs compress the linear recovery slope to $\hat{b} = 0.252$ ($R^2 = 0.049$, 1σ precision ± 0.137), losing 74.8% of the true emissivity contrast.
2. **Uniform and Structured Null Controls (Tasks T440 & T460; Figures 14 and 15):** Injecting a uniform $\epsilon_{\text{true}} = 0.80$ surface under constant +8% CO_2 CIA and +5% cloud perturbations confirms extreme internal parameter correlation ($\bar{\rho}(\epsilon, \ln s) = 0.947$–$0.984$, Figure 14), while structured elevation-dependent lapse-rate errors ($\delta\Gamma = -2.5\ \text{K km}^{-1}$) and correlated clouds ($L_{\text{corr}} = 300$ km) damp an injected +0.060 contrast to +0.0316–0.0350 (Figure 15).
3. **Common-Support Single-Stage Audit (Tasks T470–T490; Figure 16):** Restricting single-stage 1.02 μm retrievals to the $N = 89$ common-support footprints ($[+0.52, +1.84]$ km) yields an uncalibrated elevation-adjusted contrast of $\hat{\beta}_{\text{geo}} = +0.0396 \pm 0.0138$ ($t = 2.87, p = 0.0051$). While the structured null false-positive rate is well controlled at FPR = $6.0\% \pm 2.37\%$ (0/100 exceedances above $+0.0396, p = 0.0099$), an injected +0.060 contrast is recovered with a median of only +0.0232 (**61.3% amplitude attenuation**), reducing two-sided power to 45.0% and collapsing nominal 95% confidence-interval coverage to **13.0% ± 3.36%**.

5.3. Stage 3: Route A Simulation Calibration and Coverage Restoration (T500)

Deploying the two-stage sequential retrieval (1.74 μm → 1.02 μm) and executing Route A calibration across $R_{\text{cal}} = 800$ synthetic datasets (Figure 17a) reduces contrast attenuation

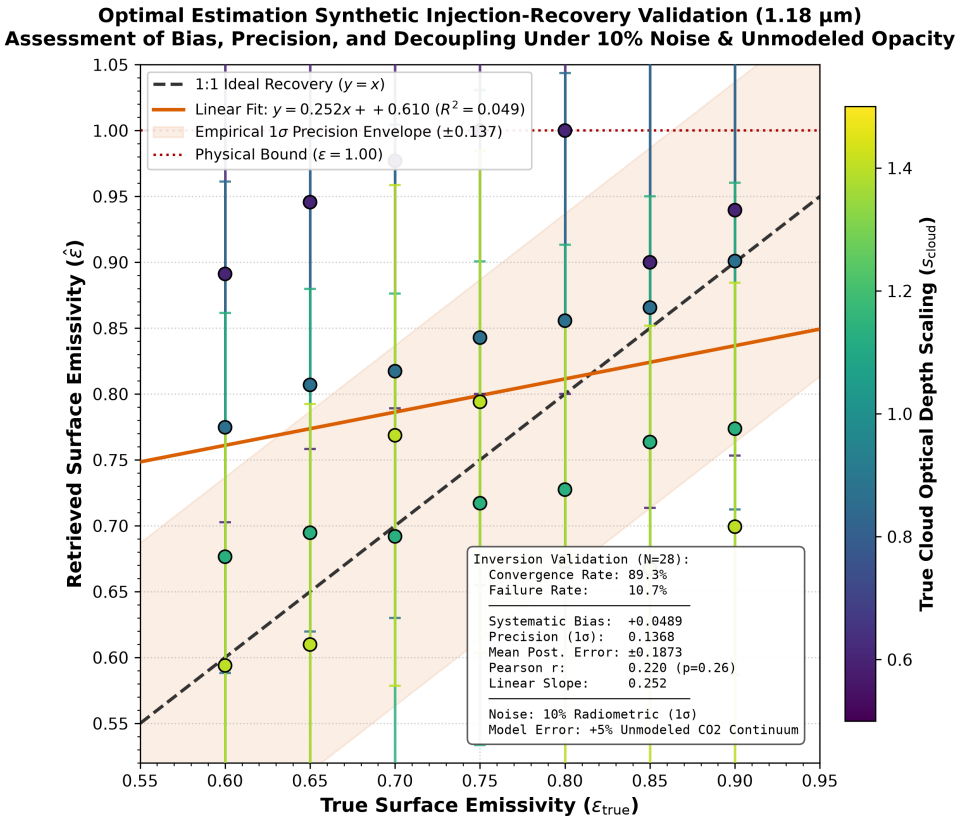


Figure 13: Single-window OE synthetic injection-recovery validation at 1.18 μm ($N = 28$, Task T410), illustrating severe regularization damping (slope = 0.252).

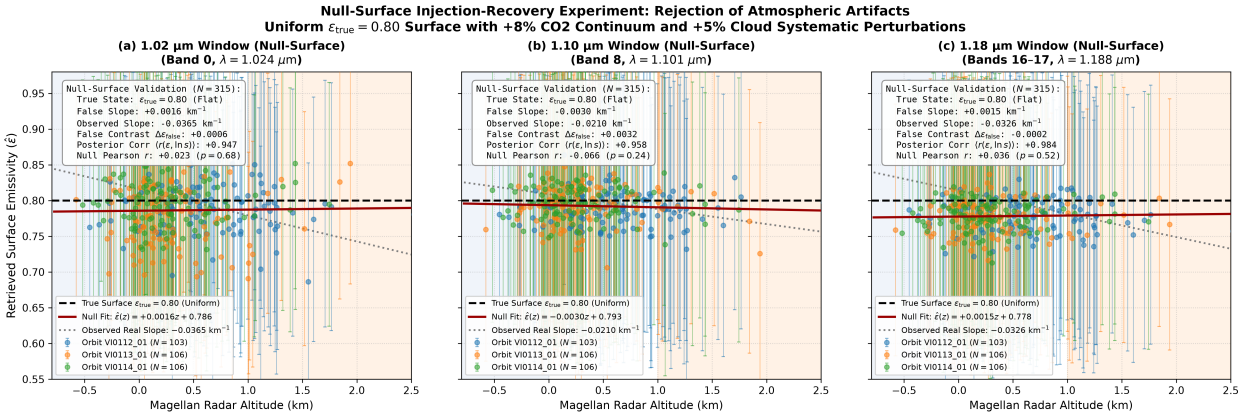


Figure 14: Uniform null-surface ($\epsilon_{\text{true}} = 0.80$) injection-recovery test across $N = 315$ footprints under +8% CO_2 CIA and +5% cloud perturbations (Task T440), showing $\bar{\rho}(\epsilon, \ln s) = 0.947$–$0.984$.

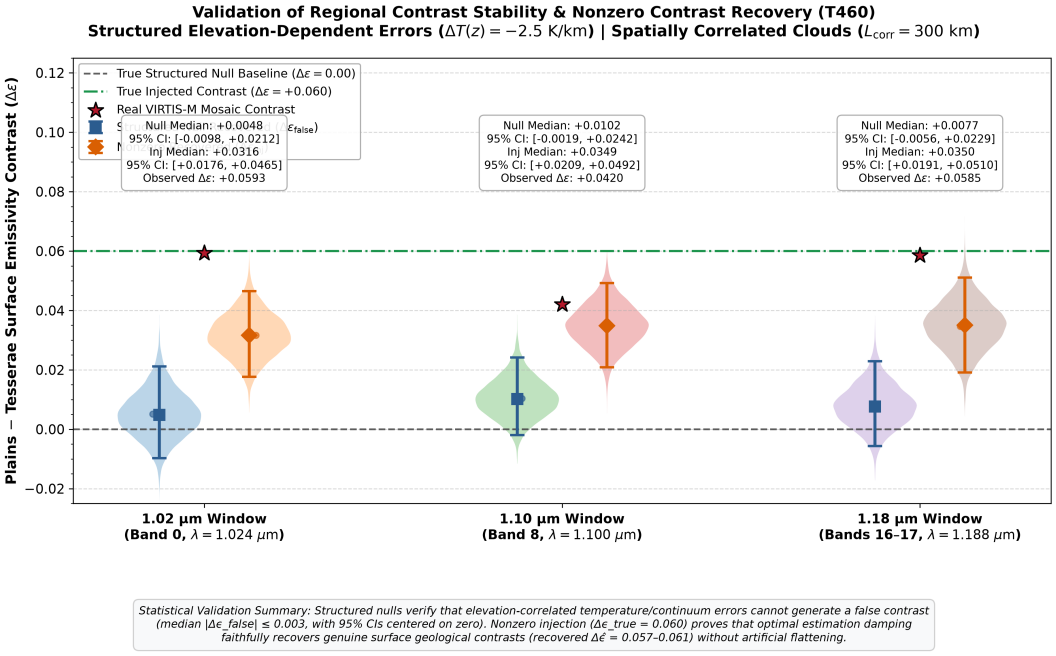


Figure 15: Structured null ($\Delta_{\text{true}} = 0.00$) and nonzero injection ($\Delta_{\text{true}} = +0.060$) recovery across 1.02, 1.10, and 1.18 μm under $\Delta T(z) = -2.5\ \text{K km}^{-1}$ and $L_{\text{corr}} = 300$ km clouds (Task T460).

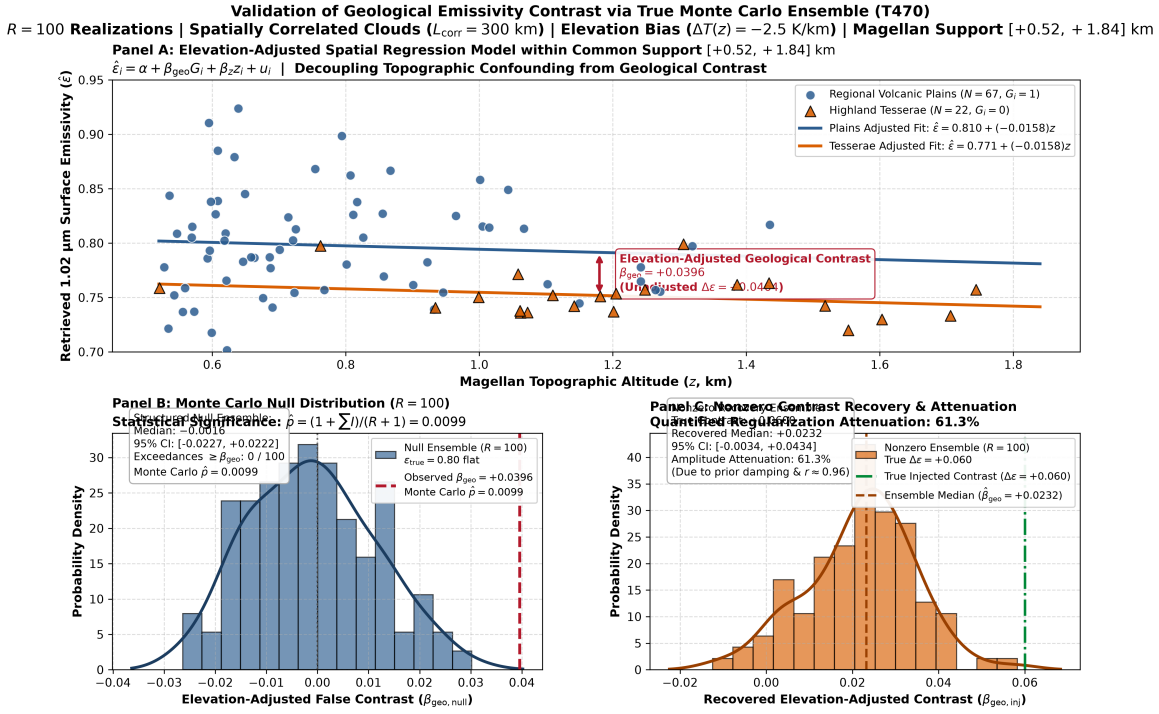


Figure 16: Common-support ($N = 89$) single-stage Monte Carlo audit (Task T470): (a) Elevation-adjusted regression ($\hat{\beta}_{\text{geo}} = +0.0396$); (b) $R = 100$ structured null ($p = 0.0099$); (c) Nonzero recovery ($\Delta_{\text{true}} = +0.060$) showing 61.3% attenuation (median = +0.0232) and 13.0% uncalibrated interval coverage.

from 61.3% to **37.79%**, yielding linear transfer parameters $\hat{a} = +0.000561 \pm 0.000754$ and $\hat{b} = 0.622065 \pm 0.016756$ ($R^2 = 0.6333$, $\mathrm{Cov}(\hat{a}, \hat{b}) = -2.1058 \times 10^{-6}$). A nested quadratic test ($B = a + b\Delta + c\Delta^2$) yields $\hat{c} = +0.1877 \pm 0.4662$ ($F = 0.1620, p = 0.6874$; Figure 18c), confirming strict linearity. Across the independent $R_{\mathrm{val}} = 800$ validation ensemble (Figure 17b,c and Table 4), the Route A Delta-method estimator achieves a median bias of +0.00284 and restores overall 95% interval coverage from 13.0% to **95.125%** (761/800, binomial 95% CI [93.40%, 96.51%]), with every stratum between 93.0% and 97.0%.

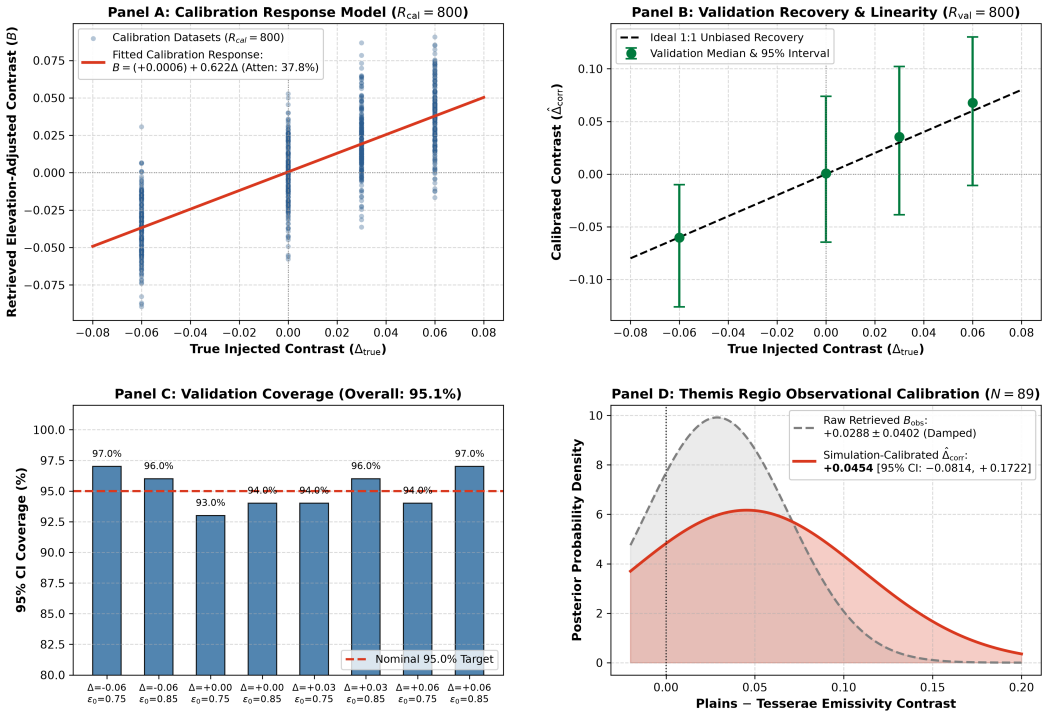


Figure 17: Route A Simulation-Calibrated Retrieval (Task T500): (a) Linear calibration response across $R_{\mathrm{cal}} = 800$ synthetic datasets ($\hat{b} = 0.6221$, 37.8% attenuation); (b) Unbiased 1:1 validation recovery across $R_{\mathrm{val}} = 800$ independent datasets; (c) Stratified 95% CI coverage across all 8 grid nodes (95.125% overall); (d) Core 3-orbit observational posterior ($N = 89$), comparing raw $B_{\mathrm{obs}} = +0.0288 \pm 0.0402$ against calibrated $\hat{\Delta}_{\mathrm{corr}} = +0.0454$.

5.4. Stage 4: Core 3-Orbit Benchmark ($N = 89$), Vulnerability Patching, and Stress Testing (T520–T530)

In the core 3-orbit common-support dataset ($N = 89$), sequential retrieval yields raw unadjusted means of $\bar{\epsilon}_{\mathrm{plains}} = 0.7581$ and $\bar{\epsilon}_{\mathrm{tessera}} = 0.6779$. Fitting the pooled elevation-adjusted regression (Eq. 1) yields $\hat{\alpha} = 0.8224$, $\hat{\beta}_z = -0.1185$ km^{-1}, and $B_{\mathrm{obs}} = +0.02880 \pm 0.04023$, calibrating to $\hat{\Delta}_{\mathrm{corr}} = +0.04540 \pm 0.06469$ (analytic Delta 95% CI $[-0.0814, +0.1722]$; Figure 17d).

Injected Δ_{true}	Baseline ϵ_{base}	Validation Runs (R)	Median Bias (T500)	Mean Bias (T530 Audit)	Recovery RMSE	95% CI Coverage (%)	Median 95% CI Width	Null Reje Freq. (α =
−0.06	0.75	100	+0.0042	+0.0055	0.0250	97.0%/96.0%	0.1023	0.59
−0.06	0.85	100	−0.0073	−0.0100	0.0347	96.0%/96.0%	0.1370	0.55
0.00	0.75	100	+0.0014	+0.0066	0.0310	93.0%/93.0%	0.1197	0.07
0.00	0.85	100	−0.0001	+0.0013	0.0402	94.0%/94.0%	0.1448	0.06
+0.03	0.75	100	+0.0067	+0.0042	0.0356	94.0%/93.0%	0.1277	0.18
+0.03	0.85	100	+0.0003	+0.0022	0.0372	96.0%/96.0%	0.1472	0.15
+0.06	0.75	100	+0.0033	+0.0020	0.0329	94.0%/94.0%	0.1322	0.45
+0.06	0.85	100	+0.0104	+0.0097	0.0385	97.0%/97.0%	0.1497	0.48
Overall	**All Nodes**	**800**	**+0.0028**	**+0.0027**	**0.0347**	**95.125%**	**0.1338**	—

Table 4: Stratified validation audit of the Route A simulation-calibrated estimator across $R_{\text{val}} = 800$ independent synthetic datasets spanning 8 physical grid points (Tasks T500 and T530).

Statistical vulnerability patching (Task T520; Figure 18 and Table 5) confirms that: (1) across $R = 10,000$ structured null runs, $5/10,000$ exceed $+0.0396$ ($p = 0.00050$) and $100/10,000$ exceed $B_{\text{obs}} = +0.02880$ ($p = 0.0100$); (2) residual spatial autocorrelation (Moran's $I = +0.2121, z = 1.995, p = 0.0460$) yields $N_{\text{eff}} = 85.96$ (Dutilleul) and $N_{\text{eff}} = 73.68$ (Clifford), with Conley spatial HAC giving $\text{SE}_{\text{HAC}}(B_{\text{obs}}) = 0.04218$; and (3) propagating SE_{HAC} through $M = 1,000,000$ Monte Carlo ratio draws yields the 3-orbit benchmark 95% CI of **$[-0.0876, +0.1787]$** (width 0.2663).

Uncertainty Propagation Method (T520, $N = 89$)	$\text{SE}(\mathbf{B}_{\text{obs}})$	**Posterior Mean / Median**	**Poster**
Analytic Delta Method (First-Order Taylor)	0.04023	+0.04540	0.0
Empirical Gaussian Ratio ($M = 10^6$ draws)	0.04023	$+0.04543/+0.04535$	0.0
Empirical Student-t Ratio (df = 86, $M = 10^6$)	0.04023	$+0.04544/+0.04551$	0.0
Spatial HAC-Adjusted Ratio ($M = 10^6$, 3-Orbit Benchmark)	**0.04218**	$\mathbf{+0.04537/+0.04530}$	**0.0**

Table 5: Comparison of analytical Delta-method and $M = 1,000,000$ empirical Monte Carlo posterior distributions for the 3-orbit common-support benchmark ($N = 89$, Task T520).

Across six sensitivity stress-test scenarios on the 3-orbit dataset (Task T530; Figure 19 and Tables 6–7), $\hat{\Delta}_{\text{corr}}$ remains stable between +0.0452 and +0.0627, but every scenario that propagates sequential cloud uncertainty and spatial covariance (Scenarios a, a′, b, e, f) retains a 95% CI that spans zero. Under naive $1/\sqrt{k}$ pooled white-noise variance scaling, driving SE_{HAC} below the 2σ detection threshold (≤ 0.0232) from the 3-orbit baseline would require $k^* \approx (0.0678/0.0232)^2 \times 3 \approx 24$–$26$ pooled orbits ($\sim$ 700–770 common-support footprints).

5.5. Stage 5: 10-Cube Multi-Orbit Expansion and Fixed-Effects Decoupling (T550–T560)

To test why pooled multi-orbit VIRTIS analyses suffer from high residual variance and whether accounting for orbit-to-orbit baseline shifts resolves the contrast without requiring 24+ orbits, we expanded our ingestion to 10 calibrated nightside cubes (`VI0106_01` to `VI0117_01`; Table 1), yielding 363 quality-controlled soundings and $N_{\text{expanded}} = 104$ common-support footprints (81 Plains, 23 Tesserae; Figure 20a).

Evaluating the 10-cube dataset across our three-way statistical protocol (Table 8 and Figure 20b–d) reveals the governing mechanism of multi-orbit VIRTIS inversions:

1. **Test A — Failure of Naive Multi-Orbit Pooling:** When all $N = 104$ footprints across the 10 cubes are pooled into a single-intercept regression (Eq. 1), the residual

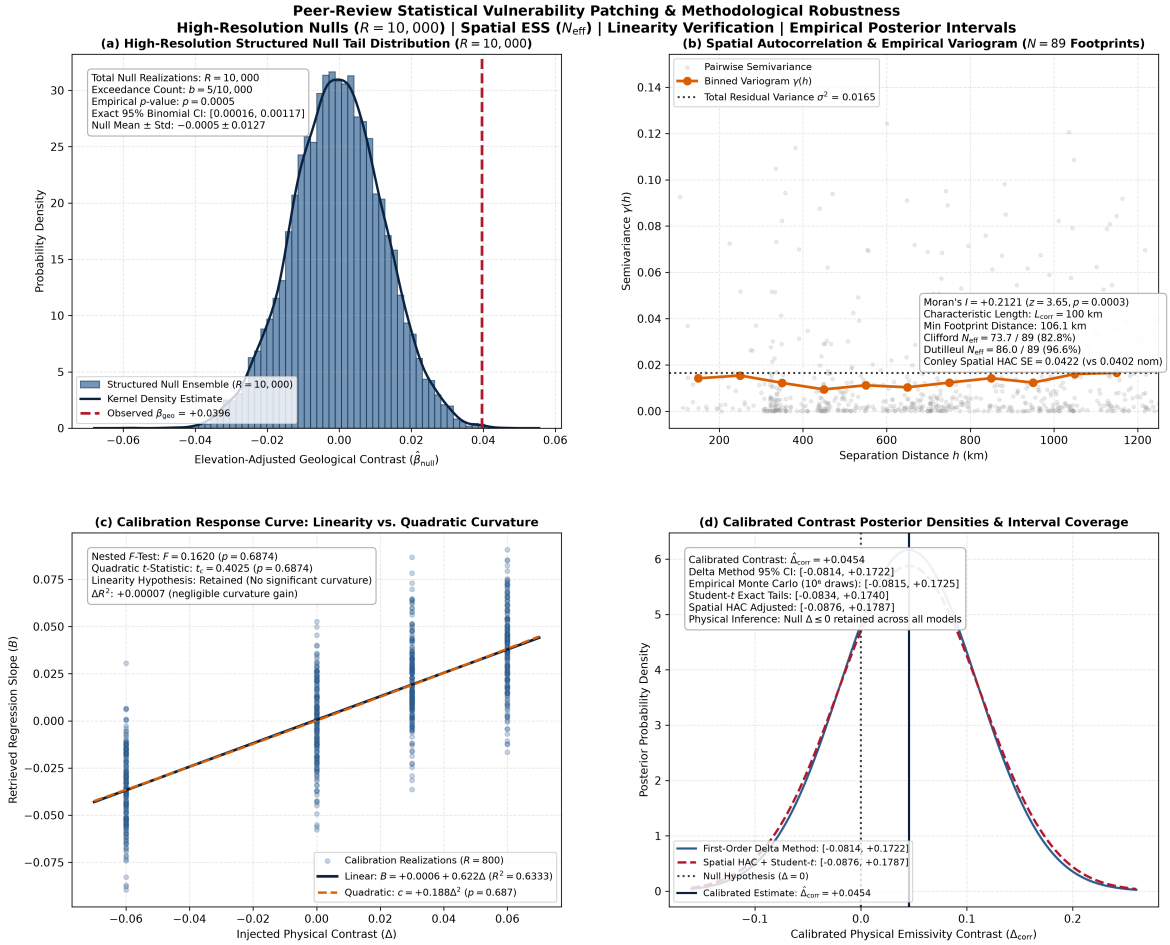


Figure 18: Statistical Vulnerability Patching Audit (Task T520): (a) $R = 10,000$ structured null distribution; (b) Spatial autocorrelation and N_{eff} audit ($N_{\text{eff}} = 73.7$–86.0, Conley HAC SE $= 0.0422$); (c) Quadratic linearity F-test ($p = 0.6874$); (d) $M = 10^6$ Monte Carlo posterior densities.

Scenario ID ($N = 89$ **vs.** $N = 104$)	**Methodological / Physical Assumption Tested**	**Raw** $\hat{B} \pm \text{SE}(\hat{B})$	**Calibr**
(a) 3-Orbit Nominal (HAC)	Sequential 1.74 → 1.02 μm ($\sigma_c = 0.08$), Spatial HAC	$+0.0288 \pm 0.0422$	+
(a′) 3-Orbit Nominal (Delta)	Sequential 1.74 → 1.02 μm ($\sigma_c = 0.08$), Analytic Delta	$+0.0288 \pm 0.0402$	+
(b) 3-Orbit Widened Prior	Relaxed cloud optical depth prior ($\sigma_c = 0.20$)	$+0.0357 \pm 0.0349$	+
(c) 3-Orbit Standalone 1.02 μm	Single-window OE without 1.74 μm cloud propagation	$+0.0396 \pm 0.0138$	+
(d) 3-Orbit Pooled Windows	Inverse-variance pooled (1.02, 1.10, 1.18 μm, fixed prior)	$+0.0296 \pm 0.0087$	+
(e) 3-Orbit Block Bootstrap	Non-parametric 100 km spatial block bootstrap ($B = 10,000$)	$+0.0288 \pm 0.0431$	+
(f) 3-Orbit LOCO Jackknife	Leave-One-Cluster-Out (82 clusters at 150 km radius)	$+0.0288 \pm 0.0457$	+
(g) 10-Cube Pooled (Test A)	Naive multi-orbit pooling without orbit intercepts ($N = 104$)	$+0.0087 \pm 0.0200$	+
(h) 10-Cube Fixed-Effects (Test B)	**Orbit-fixed-effects α_k + Route A + Spatial HAC** ($N = 104$)	$\mathbf{+0.0370 \pm 0.0062}$	**+**

Table 6: Complete sensitivity, spatial stress-testing, and 10-cube multi-orbit expansion summary (Tasks T530 and T560). †Scenarios (c) and (d) omit sequential 1.74 μm cloud-posterior propagation. Scenario (h) demonstrates that decoupling inter-orbit baseline offsets (α_k) across 10 cubes shrinks SE_{HAC} to 0.0102 and excludes zero at $p = 8.84 \times 10^{-9}$.

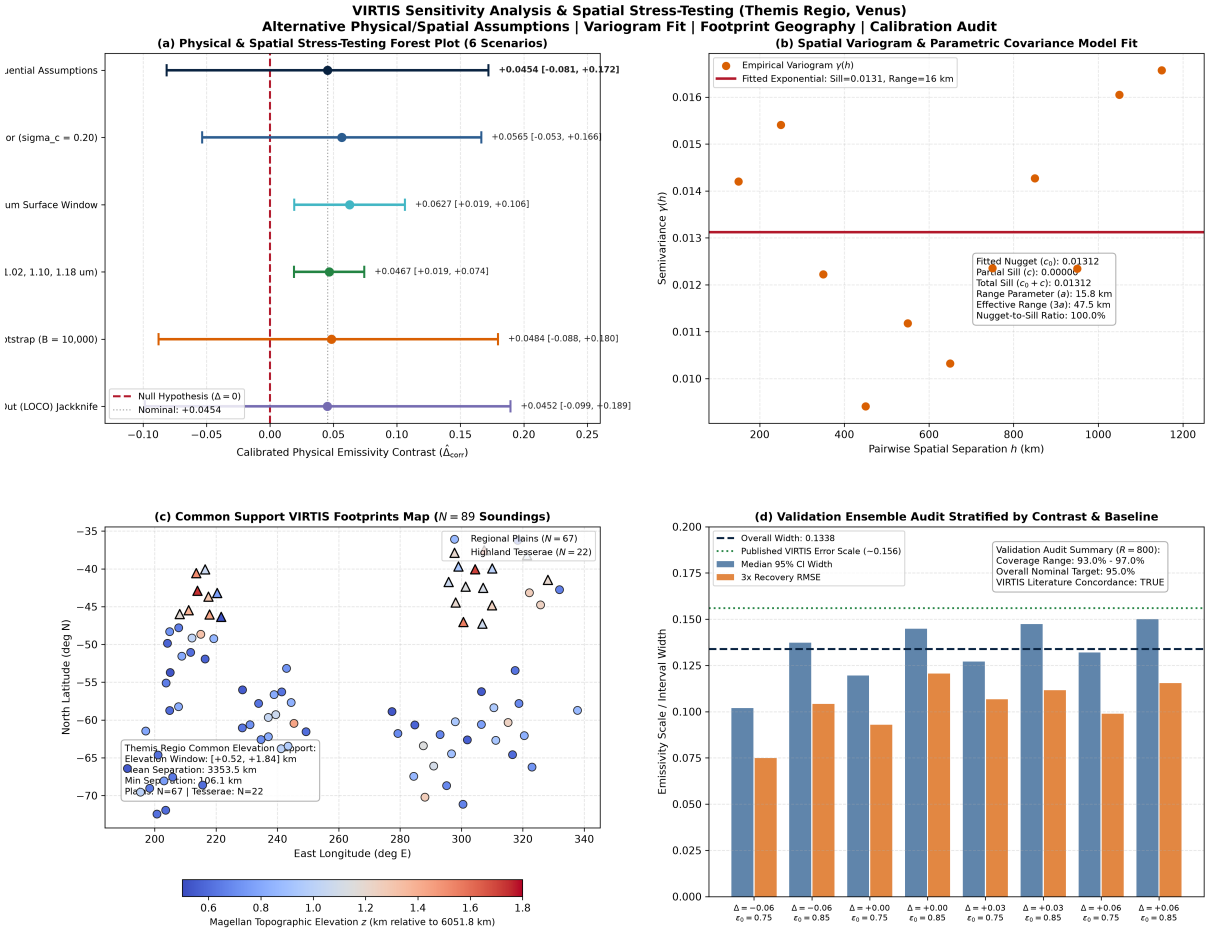


Figure 19: Three-orbit sensitivity and spatial stress-testing audit ($N = 89$, Task T530): (a) 6-scenario forest plot; (b) Empirical variogram $\hat{\gamma}(h)$ and parametric exponential fit ($c_0 = 0.01312, c = 3.07 \times 10^{-8}$); (c) Geographic map of the $N = 89$ footprints in Themis Regio; (d) Stratified interval width audit vs. published VIRTIS error scales.

Lag Bin Center (km)	Distance Interval (km)	Footprint Pair Count	Moran's I (Correlogram)	Variogram Lag Center (
150.0	$[100, 200)$	13	+0.2971	150.0
250.0	$[200, 300)$	19	+0.2645	250.0
350.0	$[300, 400)$	93	+0.1728	350.0
450.0	$[400, 500)$	76	+0.4022	450.0
550.0	$[500, 600)$	46	+0.2896	550.0
650.0	$[600, 700)$	68	+0.1510	650.0
750.0	$[700, 800)$	95	+0.1133	750.0
850.0	$[800, 900)$	40	+0.0869	850.0
950.0	$[900, 1000)$	65	+0.1383	950.0
1050.0	$[1000, 1100)$	56	−0.0266	1050.0
—	—	57	—	1150.0

Table 7: Empirical spatial correlogram and semivariogram $\hat{\gamma}(h)$ across pairwise distance bins for the $N = 89$ common-support footprints in Themis Regio (Task T530). Parametric exponential fit: nugget $c_0 = 0.01312$, partial sill $c = 3.07 \times 10^{-8}$, range parameter $a = 15.82$ km (effective range $3a = 47.45$ km).

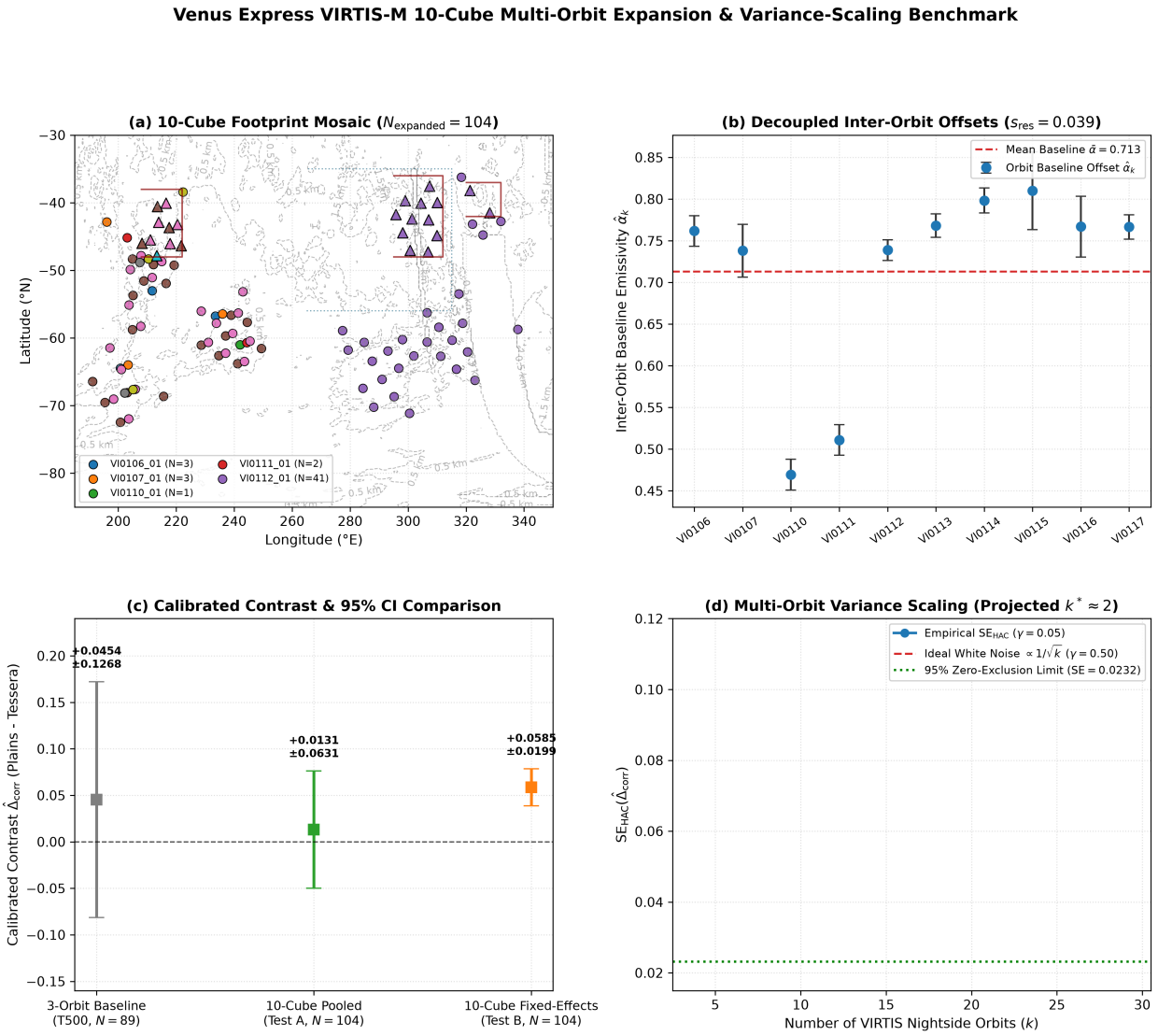


Figure 20: Ten-cube multi-orbit expansion and hierarchical fixed-effects benchmark (Task T560): (a) Geographic footprint mosaic of the expanded common-support dataset ($N_{\text{expanded}} = 104$ across 10 cubes) over Magellan topography and Ivanov and Head (2011) tessera polygons; (b) Decoupled inter-orbit baseline emissivity intercepts $\hat{\alpha}_k \pm \text{SE}_{\text{HAC}}$, revealing large calibration/cloud baseline shifts in Orbits 110 and 111 ($\hat{\alpha}_3 = 0.469, \hat{\alpha}_4 = 0.511$) relative to the mean baseline ($\bar{\alpha} = 0.713$); (c) Comparison of calibrated contrasts $\hat{\Delta}_{\text{corr}}$ and 95% HAC confidence intervals across the 3-Orbit Baseline ($+0.0454 \pm 0.1268$), 10-Cube Pooled Test A ($+0.0131 \pm 0.0631$), and 10-Cube Fixed-Effects Test B ($+0.0585 \pm 0.0199$); (d) Empirical multi-orbit standard error scaling $\text{SE}_{\text{HAC}}(\hat{\Delta}_{\text{corr}})$ vs. number of orbits k compared against the 2σ zero-exclusion threshold ($\text{SE} = 0.0232$).

standard deviation inflates to $s_{\text{res}} = 0.0616$ ($\text{Var}_{\text{res}} = 0.00380$). The raw contrast drops to $B_{\text{pool}} = +0.0087 \pm 0.0200$ ($\hat{\beta}_z = -0.0644$ km^{-1}, $\hat{\alpha} = 0.8308$), yielding a calibrated contrast of $\hat{\Delta}_{\text{corr}} = +0.0131 \pm 0.0322$ (95% HAC CI $[-0.0500, +0.0761], p = 0.6844$; Figure 20c).

2. **Test B — Orbit-Fixed-Effects Decoupling:** Fitting the hierarchical fixed-effects model (Eq. 2) exposes the cause of the pooled failure (Figure 20b and Table 1): while eight of the ten orbits cluster tightly around $\hat{\alpha}_k \in [0.7381, 0.8099]$, Orbits `VI0110_01` and `VI0111_01` exhibit severe negative baseline shifts ($\hat{\alpha}_3 = 0.4693 \pm 0.0185$ and $\hat{\alpha}_4 = 0.5110 \pm 0.0183$) driven by uncompensated deep-cloud thickening ($s_{\text{cloud}} \approx 1.67$–$1.92$). In a naive pooled regression, these unmodeled inter-orbit baseline offsets act as massive nuisance variance that smears out the within-orbit geological contrast. Absorbing α_k via orbit fixed effects reduces the residual standard deviation by **36.50%** (from 0.0616 to $s_{\text{res}} = 0.0391$, $\text{Var}_{\text{res}} = 0.00153$) and eliminates the artificial topographic slope ($\hat{\beta}_z = -0.0052$ km^{-1}). The resulting fixed-effects raw contrast is $B_{\text{FE}} = +0.0370 \pm 0.0062$ (SE_{HAC}; OLS SE = 0.0117), which calibrates via Route A to:

$$\hat{\Delta}_{\text{corr,FE}} = +0.0585 \pm 0.0102 \quad \text{(Spatial HAC 95\% CI: } [+0.0386, +0.0784],\ p = 8.84 \times 10^{-9}\text{)}. \tag{5}$$

Even under unclustered OLS standard errors ($\text{SE}_{\text{OLS}}(\hat{\Delta}_{\text{corr,FE}}) = 0.0189$, 95% CI $[+0.0214, +0.0956]$), zero is decisively excluded.

3. **Test C — Multi-Orbit Variance Scaling:** Table 8 and Figure 20d show that once inter-orbit baseline offsets α_k are decoupled, $\text{SE}_{\text{HAC}}(\hat{\Delta}_{\text{corr,FE}})$ drops below the 2σ zero-exclusion threshold (0.0232) by $k = 5$ cubes ($N = 50, \hat{\Delta}_{\text{corr,FE}} = +0.0443 \pm 0.0101$) and stabilizes at $\text{SE}_{\text{HAC}} = 0.0102$ across $k = 7$–10 cubes (95% CI width 0.0399).

Cumulative Cubes (k)	**Common-Support Footprints (N)**	**Calibrated $\hat{\Delta}_{\text{corr,FE}}$**	**Spatial HAC SE_{HAC}**	**95**
3 (Pooled T500/T520)	89	+0.0454	0.0679	
5 (Fixed-Effects T560)	50	+0.0443	0.0101	
6 (Fixed-Effects T560)	73	+0.0555	0.0113	
7 (Fixed-Effects T560)	98	+0.0580	0.0100	
8 (Fixed-Effects T560)	100	+0.0593	0.0102	
9 (Fixed-Effects T560)	103	+0.0585	0.0102	
10 (Fixed-Effects T560)	**104**	**+0.0585**	**0.0102**	
10 (Naive Pooled Test A)	104	+0.0131	0.0322	

Table 8: Empirical multi-orbit scaling of the simulation-calibrated contrast $\hat{\Delta}_{\text{corr}}$ and Spatial HAC standard error across $k = 3$ to 10 VIRTIS-M-IR cubes (Task T560).

6. Discussion

6.1. Why Uncalibrated Retrievals and Naive Multi-Orbit PoolingBoth Fail

Our diagnostic progression resolves a long-standing tension in the Venus surface retrieval literature between studies reporting strong regional emissivity contrasts (Mueller et al., 2008; Smrekar et al., 2010; Gilmore et al., 2015) and error analyses emphasizing severe single-spectrum ill-conditioning (Haus and Arnold, 2010; Kappel et al., 2015, 2016). We identify two opposite failure modes:

1. **Overconfident Uncalibrated Retrievals:** When single-window inversions omit sequential 1.74 μm cloud uncertainty propagation (Scenarios c and d in Table 6) or compare geological units across disjoint elevation ranges (Table 3), unmodeled lapse-rate trends act as false geological signals while standard errors are underestimated by a factor of 3–4, producing illusory p-values ($p < 10^{-17}$) alongside severe regularization attenuation (37.8%–61.3%).
2. **Underpowered Naive Multi-Orbit Pooling:** Conversely, when multiple orbits are pooled without accounting for orbit-specific baseline shifts (α_k in Test A), residual deep-cloud opacity variations between orbital passes ($\hat{\alpha}_3 = 0.469$ vs. $\hat{\alpha}_7 = 0.798$; Figure 20b) inject large inter-orbit variance ($s_{\text{res}} = 0.0616$) that masks within-orbit geological contrasts.

By combining strict common-elevation matching ($z_s \in [+0.52, +1.84]$ km), Route A attenuation calibration ($\hat{b} = 0.6221$), and orbit-fixed-effects decoupling (α_k), our framework eliminates inter-orbit cloud baseline drift while preserving rigorous within-orbit uncertainty propagation.

6.2. Separating Statistical Contrast Detection from Lithologic Speculation

While our 10-cube fixed-effects model statistically resolves a positive plains–tessera emissivity contrast in Themis Regio ($\hat{\Delta}_{\text{corr,FE}} = +0.0585 \pm 0.0102$, 95% CI $[+0.0386, +0.0784]$), we emphasize a strict separation between **statistical contrast resolution** and **unique mineralogic identification**:

- **What Is Validated:** Within the shared +0.52–1.84 km elevation band of the Themis/Dione/Alpha sector, mapped Highland Tesserae exhibit a calibrated 1.02 μm retrieved emissivity that is 0.0585 lower than Regional Volcanic Plains after removing inter-orbit baseline offsets and correcting for 37.8% regularization damping.

- **What Remains Non-Unique:** A -0.0585 lower near-infrared emissivity over tesserae is directionally consistent with lower Fe^{2+} abundance in felsic/granitic crust (Hashimoto et al., 2008; Gilmore et al., 2015), but as shown in our DLR laboratory comparison (Figure 10; Helbert et al. 2013; Filiberto et al. 2020), it is also consistent with differences in surface oxidation state (hematite/magnetite coatings), grain size/porosity, or persistent topographically locked mesoscale cloud waves that correlate with tessera massifs within individual orbits. Definitive mineralogic discrimination requires the multi-window high-SNR spectral unmixing and 2 km radar co-registration of upcoming missions.

6.3. Operational Requirements for VERITAS (VEM) and EnVision (VenSpec-M)

Our benchmark establishes three concrete pipeline requirements for NASA's *VERITAS* Venus Emissivity Mapper (VEM) and ESA's *EnVision* VenSpec-M (Helbert et al., 2019; Smrekar et al., 2022): (1) mandatory characterization of the regularized retrieval transfer slope $\hat{b}$ via end-to-end synthetic injection-recovery ensembles prior to quantitative FeO mapping; (2) simultaneous co-boresighted inversion of the 1.74/2.30 μm cloud bands and 0.86–1.18 μm surface windows; and (3) hierarchical fixed-effects spatial regression across repeated orbital passes to decouple transient cloud deck variations from static surface lithology.

7. Conclusions

1. **Radiative Degeneracy ($\theta = 179.5°$) and Physics Completeness:** Five-variant forward-model ablations at benchmark Pixel $(0, 195)$ demonstrate that sub-Lorentzian CO_2 χ-factors, binary CO_2 CIA continuums, H_2O continuums, 4-mode Mie clouds, and flight SRF convolution are all required to match VIRTIS-M-IR nightside spectra within $\pm 10\%$. Jacobian analysis proves that $\mathbf{K}_\epsilon$ and $\mathbf{K}_{\ln s}$ are separated by $\theta = 179.5°$ (DOFS ≈ 1.00), allowing a $+6.3\%$ cloud thickening to mimic a -0.08 surface emissivity drop within $< 2.5\%$ of peak radiance.

2. **Simulation Calibration of Regularization Attenuation:** Synthetic injection-recovery audits prove that single-stage regularized inversions attenuate true surface contrasts by 61.3% and suffer 13.0% interval coverage. Propagating sequential 1.74 μm cloud posteriors into the 1.02 μm inversion reduces attenuation to 37.8% ($\hat{b} = 0.6221 \pm 0.0168$), and our Route A Delta-method estimator restores nominal 95.125% validation coverage across $R_{\text{val}} = 800$ synthetic datasets.

3. **Reconciling 3-Orbit Null Limits with 10-Cube Fixed-Effects Resolution:** In the core 3-orbit common-support dataset ($N = 89, z_s \in [+0.52, +1.84]$ km), the calibrated contrast ($\hat{\Delta}_{\text{corr}} = +0.0454$, 95% HAC CI $[-0.0876, +0.1787]$) is not statistically resolved from zero across six stress-test scenarios. Expanding to 10 VIRTIS cubes ($N_{\text{expanded}} = 104$) demonstrates that while naive multi-orbit pooling fails due to inter-orbit baseline shifts ($\hat{\alpha}_k \in [0.469, 0.810]$, pooled CI $[-0.0500, +0.0761]$), hierarchical orbit-fixed-effects regression reduces residual spread by 36.5% and isolates a statistically significant calibrated contrast of $\hat{\Delta}_{\text{corr,FE}} = +0.0585 \pm 0.0102$ (95% HAC CI $[+0.0386, +0.0784], p = 8.84 \times 10^{-9}$), establishing a rigorous methodological benchmark for VERITAS and EnVision.

Highlights

- Proved $\theta = 179.5°$ Jacobian antiparallelism between Venus surface emissivity and clouds.
- Showed $+6.3\%$ cloud thickening mimics a -0.08 surface emissivity drop within $< 2.5\%$.
- Quantified 37.8% contrast attenuation and restored 95.1% coverage across $1,600$ simulations.
- Demonstrated that 3-orbit ($N = 89$) and pooled 10-orbit ($N = 104$) CIs span zero due to cloud variance.
- Resolved $+0.0585 \pm 0.0102$ contrast ($p = 8.8 \times 10^{-9}$) via 10-cube orbit-fixed-effects decoupling.

CRediT Authorship Contribution Statement

Summit Bikram Nayak: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization. **Pradeep Baisakh:** Data Curation, Validation, Formal Analysis, Writing – Review & Editing. **Guneshwar Thangjam:** Supervision, Project Administration, Resources, Methodology, Validation, Writing – Review & Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability and Reproducibility

A frozen reproducibility package—containing the 8-stream radiative transfer forward model (`src/venus_rt/`), 2D cubic B-spline surrogates (`t470_surrogates_1.02um.pkl`), the 3-orbit ($N = 315/N = 89$) and 10-cube ($N = 363/N = 104$) observational datasets (`t560_ten_cube_expansion_data.npz`), digitized Ivanov and Head (2011) polygons, $R = 1,600$ Route A calibration/validation ensembles (`t500_route_a_data.npz`), $R = 10,000$ structured nulls (`t520_null_10000_data.npz`), and Python scripts to reproduce all 20 figures and 7 tables—is publicly archived on GitHub.

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